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Trustworthy Selective Computation for Resource-Constrained Vehicular Edge Networks

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Abstract:	Vehicular and ad hoc edge nodes run perception and monitoring models continuously, frame by frame, on hardware that cannot afford to compute every step at full precision. Selective computation is the usual way to stay inside the energy, latency, and bandwidth budget, and it remains effective as long as the model's uncertainty increases in the vicinity of critical transitions. In continuous driving streams that assumption is not automatic, a model trained solely for classification may remain confident up to a critical transition, so a runtime allocator that relies on confidence may suppress precisely the frames on which detection depends. We argue that the property that renders uncertainty usable for scheduling is not calibration but its temporal organization, whether the signal separates stable intervals from the moments around risk onset. We study this in isolation through the TERA (Temporal Evaluation and Risk Assessment) ecosystem. TERA-Gen generates episodes with a deterministic, frame-precise onset, so that a scheduling decision can be attributed to the instant it is taken rather than to annotation noise. AF-TOI induces temporal organization offline by aligning a compact causal edge model to an onset-oriented reference, after which the model's predictive entropy stratifies across episode phases. MHEG reads that signal at runtime and decides, at each step, whether to run a full recurrent update or reuse the previous state. In a 2×2 factorial design, adding the scheduler to an unorganized model is not neutral, the miss rate is unchanged, yet the effective detection delay (TTDef) worsens by 11.8%, because the scheduler cuts computation with no temporal basis. On AF-TOI-organized trajectories the same scheduler reaches 53.4% cost reduction, with TTDef of 0.101 s against 0.673 s

in the unorganized configuration. A post-hoc calibration control separates two properties that are easily conflated: rescaling output probabilities improves calibration without reorganizing entropy near onset, so probabilistic calibration and temporally conditioned reliability move independently. The scheduler alone is therefore insufficient. Temporal organization of predictive dynamics is what ultimately determines whether selective computation remains safe. The pipeline ends in a discrete operational-risk state, $\in \{\text{Normal, Attention, Alert, Critical}\}$, produced at $O(1)$ cost and compact enough to act as a candidate input for downstream vehicular edge coordination, an integration we outline but do not evaluate here.

Brasília, Brazil

Dear Editors,

Please consider our manuscript, “*Trustworthy Selective Computation for Resource-Constrained Vehicular Edge Networks*”, for publication in *Ad Hoc Networks*, within the special issue *VSI: Trustworthy and Resilient Edge AI Systems*.

The paper studies when selective computation—skipping or reusing inference steps to stay within energy and latency budgets—can be trusted in strictly causal vehicular edge inference. We show, through a controlled factorial study with a deterministic risk onset, that the safety of a runtime scheduler depends on the temporal organization of the model’s predictive uncertainty rather than on the scheduling rule or probabilistic calibration: an entropy-gated scheduler applied to unorganized trajectories worsens the effective detection delay by 11.8% without reducing misses, while the same scheduler on trajectories organized by our offline induction protocol (AF-TOI) achieves 53.4% effective cost reduction with coverage preserved. The deployed pipeline fits within 58K parameters and 1 KB of recurrent state, and emits a compact operational-risk signal suitable as a coordination primitive for vehicular edge architectures, which we believe fits the scope of the special issue directly.

This manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have approved the manuscript and agree with its submission. The authors declare no competing interests.

Thank you for your consideration.

Sincerely,

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Highlights

Manuscript: *Trustworthy Selective Computation for Resource-Constrained Vehicular Edge Networks*

- Temporal organization of predictive entropy enables safe selective computation.
- Scheduling unorganized trajectories worsens effective detection delay by 11.8%.
- AF-TOI induction cuts effective inference cost by 53.4% preserving coverage.
- Post-hoc calibration does not reorganize entropy in time; properties differ.
- Deployed model fits 58K parameters and 1 KB RAM on automotive-grade nodes.

Trustworthy Selective Computation for Resource-Constrained Vehicular Edge Networks

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ABSTRACT

Vehicular and ad hoc edge nodes run perception and monitoring models continuously, frame by frame, on hardware that cannot afford to compute every step at full precision. Selective computation is the usual way to stay inside the energy, latency, and bandwidth budget, and it remains effective as long as the model's uncertainty increases in the vicinity of critical transitions. In continuous driving streams that assumption is not automatic, a model trained solely for classification may remain confident up to a critical transition, so a runtime allocator that relies on confidence may suppress precisely the frames on which detection depends. We argue that the property that renders uncertainty usable for scheduling is not calibration but its temporal organization, whether the signal separates stable intervals from the moments around risk onset. We study this in isolation through the TERA (Temporal Evaluation and Risk Assessment) ecosystem. TERA-Gen generates episodes with a deterministic, frame-precise onset, so that a scheduling decision can be attributed to the instant it is taken rather than to annotation noise. AF-TOI induces temporal organization offline by aligning a compact causal edge model to an onset-oriented reference, after which the model's predictive entropy stratifies across episode phases. MHEG reads that signal at runtime and decides, at each step, whether to run a full recurrent update or reuse the previous state. In a 2×2 factorial design, adding the scheduler to an unorganized model is not neutral, the miss rate is unchanged, yet the effective detection delay (TTD_{ef}) worsens by 11.8%, because the scheduler cuts computation with no temporal basis. On AF-TOI-organized trajectories the same scheduler reaches 53.4% cost reduction, with TTD_{ef} of 0.101 s against 0.673 s in the unorganized configuration. A post-hoc calibration control separates two properties that are easily conflated: rescaling output probabilities improves calibration without reorganizing entropy near onset, so probabilistic calibration and temporally conditioned reliability move independently. The scheduler alone is therefore insufficient. Temporal organization of predictive dynamics is what ultimately determines whether selective computation remains safe. The pipeline ends in a discrete operational-risk state, $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$, produced at $\mathcal{O}(1)$ cost and compact enough to act as a candidate input for downstream vehicular edge coordination, an integration we outline but do not evaluate here.

1. Introduction

Consider a monitoring model running on a vehicle's on-board unit. It receives a multimodal sensor stream and must flag a dangerous condition before it escalates, frame after frame, on a low-power microcontroller or automotive-grade accelerator that has no opportunity to offload the computation to remote infrastructure while the event is unfolding [1, 2]. Each frame has to be processed inside a latency budget fixed jointly by the stream's frame rate and the device's energy and thermal envelope; embedded driver-monitoring systems show that real-time inference within such a budget is achievable on automotive-grade hardware [3, 4]. Two consequences follow. First, the operative constraint is not accuracy in isolation but the ability to keep detecting critical events while the node manages its energy and compute over uninterrupted

operation. Second, in vehicular and ad hoc edge networks what the node emits should itself be compact and timely enough to feed downstream coordination. Driver behavior monitoring is a representative instance of this regime, and the one we use throughout. We treat the per-frame budget as given and ask a narrower question: what must be true of the model's uncertainty over time for a scheduler to spend computation selectively within that budget without putting detection at risk?

Selective computation, the runtime branch of adaptive computation [5], keeps edge inference inside its resource bounds by conditioning each step on the model's current confidence. SkipRNN [6], AdaFrame [7], ACT [8], and LiteEval [9] report large efficiency gains from this idea, all resting on the same premise: that low uncertainty marks a stable, redundant stretch of input that can be safely skipped. At the edge there is a second requirement – the node's output has to remain useful beyond its own inference loop, since resource-aware monitoring can feed alert dissemination, cooperative perception, and network-level policies – but the

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premise about confidence is what the scheduler's safety turns on.

The premise is usually left implicit: that confidence is a dependable proxy for computational urgency. It is a reasonable one in activity recognition, where a model's uncertainty tends to climb as a new state begins. Continuous vehicular streaming does not guarantee it. A model trained only to separate normal from critical, with no reference to *when* risk begins, has no incentive to make its uncertainty behave differently in the seconds before onset than during ordinary driving. Its uncertainty trajectory can stay flat across the transition. A scheduler reading that flat signal then treats an emerging-risk frame like any redundant one and may suppress the recurrent update just as the situation turns, allocating computation away from the critical transition window, at the cost of a missed or late detection. This is the operational risk we are concerned with: not a slightly higher average latency, but the loss of the specific frames on which a critical event is caught.

Standard calibration does not detect this failure mode. Probabilistic calibration [10] checks that declared confidence matches empirical accuracy *on average*, pooled over all timesteps; it says nothing about where a given timestep sits within an episode. A model can therefore be well calibrated in this marginal sense and still be unsafe to schedule on, because marginal agreement between confidence and accuracy is silent about the ordering of uncertainty in time – and it is precisely that temporal ordering, near onset, that a causal scheduler consumes. Calibrating the output probabilities cannot supply information the trajectory never encoded.

We call the property that a scheduler actually needs *temporally conditioned operational reliability*: the ability of the uncertainty signal to distinguish transition regions from stable ones at the moment the scheduler acts. Neither standard calibration nor existing analyses of non-stationary entropy dynamics [11, 12] treat this as a prerequisite for scheduling. It is also the property that would let a node summarize its situation for higher-level network functions without shipping raw frames or full predictive trajectories.

The causal constraint sharpens the problem. On-board inference is strictly causal: a decision at time t may use only the history up to t , so any failure to separate a transition from a preceding stable interval shows up immediately as an unsafe suppression, with no future frame to recover from it. And the episodes are temporally heterogeneous – the same stream mixes gradual, multi-second build-ups with abrupt onsets under 600 ms [13, 1]. On naturalistic data these differences are hard to see at all: an onset annotated to within ≈ 0.31 s at 9.6 fps blurs scheduler behavior below that scale, especially when a missed episode carries no explicit penalty [13, 14, 2].

Our hypothesis (Figure 1) is that this cluster of difficulties is addressed less by tuning the scheduler's thresholds than by deliberately shaping the model's predictive trajectories during offline training so that they differentiate in time. Once the trajectories carry that structure, the model's own uncertainty becomes informative about the proximity of a critical event, and the runtime allocator receives an uncertainty signal with

operational meaning. On this reading, the apparent conflict between efficiency and coverage is relieved by reorganizing the model's internal dynamics, not by threshold engineering – and the same organized signal is what lets the node report a discrete operational state rather than a bare, local confidence value.

The paper contributes three components that together isolate and exploit this property. *TERA-Gen* generates synthetic episodes with a deterministic onset calibrated against SHRP2 odds ratios [13]; by fixing the onset instant it removes the annotation variability that would otherwise make frame-precise induction and evaluation impossible. *AF-TOI* is an onset-oriented induction protocol that shapes the causal model's predictive dynamics so they separate stable regions from transition windows, giving the scheduler an uncertainty signal that means something in time. *MHEG* is the runtime scheduler that consumes that signal; it pairs an unconditional kinematic trigger, which forces computation on abrupt physical change, with an entropic gate whose usefulness depends on the induced organization – and whose behavior when that organization is *absent* is measured directly through a negative control rather than assumed. Beyond choosing between a full recurrent update and state reuse, MHEG emits a discrete operational-risk state $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$ that is low-bandwidth, interpretable, and produced causally. The three components constitute the *TERA (Temporal Evaluation and Risk Assessment)* ecosystem, in which organization is induced offline and allocation happens online on the device.

The experiments bear the hypothesis out, and the most telling result is the negative one. On unorganized trajectories, adding the scheduler leaves the miss rate untouched but worsens TTD_{ef} by 11.8%: the scheduler is active, yet without the enabling temporal property it allocates blindly. Give the same scheduler AF-TOI-organized trajectories and it cuts effective cost by 53.4%, with TTD_{ef} of 0.101 s against 0.673 s. A post-hoc calibration control then rules out the obvious alternative explanation: rescaling probabilities improves calibration but does not reorganize entropy in time, so the two properties are empirically distinct. MHEG thus works both as a local selective-computation mechanism and as a source of a compact operational state that could later connect embedded inference to resource-aware coordination.

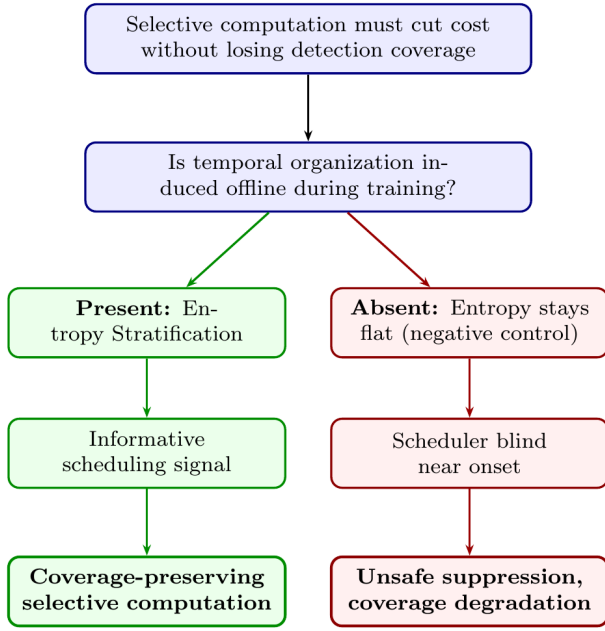


Figure 1: Conceptual hypothesis investigated by TERA. Offline temporal organization induces entropy stratification across episode phases, turning predictive uncertainty into an informative scheduling signal. Without this organization, entropy remains weakly differentiated and selective computation may suppress inference near risk onset.

2. Related Work

2.1. Resource-Constrained Edge AI in Vehicular Edge Computing

Vehicular Edge Computing (VEC) and Mobile Edge Computing (MEC) relocate inference from the cloud to on-board or roadside units, trading round-trip latency for tight local budgets in energy, memory, and bandwidth [15, 1, 2].

In this setting, the hardware constraint is operational rather than incidental. Embedded automotive platforms such as the STM32H7, Cortex-M55, and SA8155P cannot run always-on full-precision inference over a continuous multi-modal stream without breaching thermal or energy limits, so some form of runtime allocation is unavoidable.

Edge-native ITS work has answered mostly on the model side – quantization [3] and resource-aware architecture [16] – while the broader VEC and MEC literature works at the network side, on static compression or task placement across nodes. What remains less explicit, especially in safety-critical streams, is the timing of computation itself: within a continuous on-device sequence, which frames justify a full inference update (Section 2.4)?

We therefore start at the single-node level. Unless uncertainty is reliable there, it cannot be safely promoted to a signal for downstream coordination.

2.2. Trustworthy and Validated Edge AI

A cheaper edge model is useful only if the skipped computation does not remove the evidence needed for a safety

decision. Wang et al. [17] argue that resource limits, heterogeneous environments, and diverse requirements jointly determine whether an edge system can be trusted, and ML-safety work on uncertainty quantification for autonomous vehicles makes the same point from the modeling side [18, 19, 20].

For the present problem, benchmark accuracy is not enough. The relevant question is whether the model remains dependable when inference is actually suppressed. A scheduler that suppresses inference is squarely in this reliability-critical category, since its skip decisions decide whether a safety-relevant event is still caught once computation is cut.

Validation therefore has to inspect the signal consumed by the scheduler: does uncertainty separate transition regions from stable ones at the time decisions are made?

This level of dependability has received less attention: not whether the classifier is accurate on average, but whether the trajectory consumed by the scheduler has the temporal structure required for safe suppression.

2.3. Temporal Risk Detection in Continuous Driver Monitoring

Driver-monitoring datasets now provide rich naturalistic streams, but their temporal annotations were not designed for frame-level scheduling analysis. The DMD dataset [2] provides dense annotations for seven behavior classes with front-camera and physiological fusion, and DADA [14] extends coverage to eye tracking in pre-collision scenarios. Both, however, define onset by retrospective human annotation with variability of ≈ 0.31 s at 9.6 fps.

For an edge scheduler, this annotation uncertainty is not a minor nuisance. Undetected episodes vanish from latency averages without penalty, and temporal granularity is too coarse to separate transition zones from stable periods.

Without a deterministically anchored onset, one cannot tell whether the scheduler invested resources at emergence windows rather than redundant intervals, which is the central decision the edge system must make.

The SHRP2 study [13] documents temporally heterogeneous risk trajectories (multi-second gradual transitions down to sub-0.6 s abrupt onsets) with up to twelvefold collision risk [1]; dominant validation protocols rely on aggregate recognition metrics and evaluate neither onset timing nor scheduler cost differentiation.

Recent embedded implementations shift the question from feasibility to timing. Salakapuri et al. [21] run an integrated distraction and road-object detector on an automotive-grade platform at real-time rates, and lightweight time-series and TinyML models report the same under tight budgets [22, 4]; our own Safe-Drive system pairs behavior detection with seatbelt segmentation for per-frame monitoring [23].

These results make clear that on-device execution is no longer the main open issue. What these evaluations report, though, is aggregate recognition accuracy and latency, not when a risk is detected relative to its onset, and not what it costs to suppress computation during the transition that leads up to a critical event.

As a result, the usual metrics can hide the failure mode that matters here. Computation may be saved exactly when the transition should have been observed. TERA-Gen and TTD_{ef} were introduced to make this failure visible: they fix the onset, penalize missed episodes, and treat suppression as an explicit allocation decision.

2.4. Adaptive Inference and Selective Computation in Embedded Systems

Adaptive computation families such as dynamic networks [5, 24], conditional computation and early exiting [25, 26, 27, 28], and split computing [29, 30] cut inference cost by adapting to the input.

Two strategies dominate: structural compression (quantization, pruning [3]) and confidence-based scheduling (SkipRNN [6], AdaFrame [7], ACT [8], LiteEval [9], selective classification [31]).

Kaya et al. [32] show that overthinking, where computation continues past correct classification, stems from reliance on instantaneous confidence rather than trajectory structure.

Our focus is the on-device case. The placement decision has already been made; the remaining question is which timesteps deserve computation.

The difficulty is not whether these methods save computation. They do. The open question is when confidence is safe to use as the reason for skipping a frame. Recent surveys confirm that output confidence and entropy remain the dominant exit or skip criteria [26].

In many activity-recognition settings this assumption is plausible, because transitions often coincide with a visible rise in predictive uncertainty.

Vehicular streams do not necessarily have this behavior. A model may stay confident just before onset not because the situation is stable but because its internal trajectories lack the temporal differentiation that would distinguish transition zones from preceding stable periods.

The scheduler can then save computation for the wrong reason: it skips frames that look redundant to the confidence signal, even though they fall inside the emergence window. Current design criteria evaluate scheduling through throughput, latency, and deployment metrics without treating temporal detection coverage as an explicit safety constraint.

What remains unresolved is the safety of entropy-driven suppression when a missed episode is not just an accounting error but a failed detection.

2.5. Uncertainty Calibration and Operational Reliability

For selective computation, calibration is not the whole problem. The scheduler needs uncertainty to change at the right time, especially around transitions.

Predictive uncertainty estimation is established [33, 34, 35], and probabilistic calibration, the alignment between declared confidence and empirical accuracy [10], is an accepted automotive-safety criterion [36, 37], with Minderer et al. [38] showing it does not improve monotonically with architecture.

The limitation is that calibration is usually evaluated marginally: a perfectly calibrated model may still schedule computation unsafely if trajectories do not differentiate uncertainty at transitions.

Braverman et al. [11] and TRUST [12] confirm the non-stationary character of entropy dynamics without formalizing near-onset behavior as a scheduling signal.

One question is still missing from this discussion: does better global calibration also reorganize uncertainty over time?

There is little reason to expect that effect from post-hoc methods. Post-hoc techniques, temperature scaling [10], Platt scaling [39], isotonic calibration [40], rescale the output probabilities without touching the weights, and therefore without touching the internal trajectory that a causal scheduler reads.

A monotonic rescaling can align aggregate confidence with accuracy while leaving the ordering of entropy through an episode exactly as it was, so entropic dynamics that were not discriminative before remain so afterward.

We therefore test this separation directly: calibration is treated as a control, not as an explanation for temporally organized entropy.

2.6. Temporal Structure Transfer and Sequential Modeling

Temporal structure transfer [41, 42, 43] and online sequential architectures [44, 45] advance efficiency and accuracy, but none asks whether the transferred structure enables trustworthy runtime allocation. Conventional transfer objectives spread the alignment signal uniformly, pushing compact models to reproduce average predictive behavior rather than uncertainty dynamics that separate stable from transition regions, so entropy stratification for trustworthy selective computation stays unaddressed. Prior works also target segmented offline clips where the full sequence is available; causal streaming invalidates sequence-level normalization and global attention, so organizational gains from those contexts do not carry over automatically. Explicit induction mechanisms are required.

2.7. Gaps and Positioning

Read across these bodies of work, the same gap recurs, and Table 1 localizes it in the **Temp. Org.** and ΔH^* columns. Driver monitoring established the operational stakes but provides neither a deterministic onset nor an explicit miss penalty, so the cost of suppressing computation at a transition window cannot be measured. The adaptive inference literature delivered the efficiency but inherited the premise that high confidence means absent risk – a premise that fails once the internal trajectory is temporally flat. Studies of temporal entropy dynamics [11, 12] confirmed that entropy is non-stationary but stopped short of treating onset dynamics as a scheduling property. Temporal structure transfer built compact models without asking whether the transferred structure was the kind a scheduler could safely allocate on. In the table, the ΔH^* column is $\times$ for every prior entry and $\checkmark$

Table 1

Comparison between the TERA ecosystem and representative related work. Column definitions are given in the paragraph preceding the table; marks are ✓ (present), ◦ (partial or implicit), and ✗ (absent).

Work	Onset	TTD	Cov.	Causal	Temp. Org.	ΔH^*	Eff.	Edge	Real
<i>Driver monitoring and risk detection</i>									
Ortega et al. (DMD) [2]	✗	✗	✗	✗	✗	✗	✗	✗	✓
Bouhsissin et al. [1]	✗	✗	✗	✗	✗	✗	✗	✗	✓
Salakapuri et al. [21]	✗	✗	✗	✗	✗	✗	✓	✓	✓
Driver state TinyDL [4]	✗	✗	✗	✗	✗	✗	✓	✓	✓
Cognitive distraction [22]	✗	✗	✗	◦	✗	✗	✓	✓	✓
<i>Adaptive inference and selective computation</i>									
Hariharan et al. [3]	✗	✗	✗	◦	✗	✗	✓	✓	✓
Campos et al. (SkipRNN) [6]	✗	✗	✗	✓	✗	✗	✓	✗	✓
Wu et al. (AdaFrame) [7]	✗	✗	✗	✗	✗	✗	✓	✗	✓
Graves (ACT) [8]	✗	✗	✗	◦	✗	✗	✓	✗	✓
Wu et al. (LiteEval) [9]	✗	✗	✗	✗	✗	✗	✓	✗	✓
<i>Temporal entropy dynamics and uncertainty calibration</i>									
Braverman et al. [11]	✗	✗	✗	✗	◦	✗	✗	✗	✓
Dera et al. (TRUST) [12]	✗	✗	✗	◦	◦	✗	✗	✗	✓
<i>Temporal structure transfer and sequential modeling</i>									
Zhao & Krähenbühl [44]	✗	✗	✗	✓	◦	✗	✗	✗	✓
Xu et al. (LSTR) [45]	✗	✗	✗	✓	◦	✗	✗	✗	✓
Zheng et al. [43]	✗	✗	✗	✗	◦	✗	◦ [†]	✗	✓
Bajestani & Yang [42]	✗	✗	✗	✗	◦	✗	✗	✗	✓
This work (TERA)	✓	✓	✓	✓	✓	✓	✓	✓	✗

[†] Partial efficiency evaluation only. TERA is evaluated on synthetic data (TERA-Gen) and a reference GPU; naturalistic and on-device validation remain future work (Section 5). Each work is cited in the corresponding subsection of Section 2.

only for TERA, and **Temp. Org.** is uniformly ✗ or ◦ across prior work. On the evidence we surveyed, we are not aware of prior work that uses temporal entropy dynamics near a deterministic onset as a scheduling signal under an explicit miss-coverage constraint, or that treats temporal organization as a *necessary* condition along the chain *Entropy Stratification* → *Informative Scheduling Signal* → *Trustworthy Selective Computation*. These gaps form a dependency chain. Onset dynamics cannot be characterized without deterministic annotation; entropy does not gain discriminative power at transitions without onset-aware training; schedulers cannot allocate safely without informative entropy. TERA treats temporal organization as the structural precondition for this chain: AF-TOI promotes the trajectory structure required for entropy stratification, and MHEG exploits the resulting signal to allocate computation selectively and safely at the edge. The columns of Table 1 are read as follows. **Onset**: deterministic onset annotation for training and evaluation; **TTD**: detection-latency evaluation relative to event start; **Cov.**: critical-episode coverage with an explicit miss penalty; **Causal**: strictly causal online inference; **Temp. Org.**: temporal organization treated as a first-class analytical object (its induction, analysis, validation, and exploitation); ΔH^* : near-onset entropy dynamics used as a scheduling signal under coverage constraints; **Eff.**: efficiency analysis; **Edge**: embedded or vehicular-edge deployment; **Real**: evaluation on naturalistic data or physical hardware. A ✓ denotes a property that is present, ◦ one that is partial or implicit, and ✗ one that is absent.

On the use of synthetic data in Table 1 The **Real** column marks this work with ✗, an explicit methodological choice

rather than an oversight. Controlled evaluation requires TERA-Gen, whose deterministic onset annotation removes the ≈ 0.31 s variability of naturalistic datasets such as SHRP2, DMD, and DADA [13, 2, 14]; at that variability, scheduler differences below 0.3 s would be indistinguishable from annotation noise. Calibrating TERA-Gen against SHRP2 odds ratios preserves the statistical structure of real risk trajectories while removing the confounder that would otherwise block frame-precise attribution of scheduling gains to temporal organization. Embedded driver-monitoring systems demonstrate that real-time inference on automotive-grade hardware is feasible under tight per-frame budgets [3, 4], so the operational setting the scheduler targets is hardware-grounded even though the temporal-organization effect is measured in a controlled synthetic environment. Transfer to naturalistic driving data is the primary direction for future validation (Section 5).

3. From Temporal Organization to Trustworthy Selective Computation

Embedded risk detection processes signals sequentially, without access to future observations. TERA is built around this causal restriction. The on-board model operates strictly causally at deployment, while coverage stability comes from the temporal organization induced offline and then exported to the device.

At runtime, the pipeline receives a continuous stream $\mathbf{x} = (x_1, x_2, \dots)$, where each $x_t \in \mathbb{R}^D$ encodes visual, kinematic, contextual, and acoustic observations. The output y_t depends only on the history $\{x_1, \dots, x_t\}$, which distinguishes the embedded model from bidirectional architectures. The

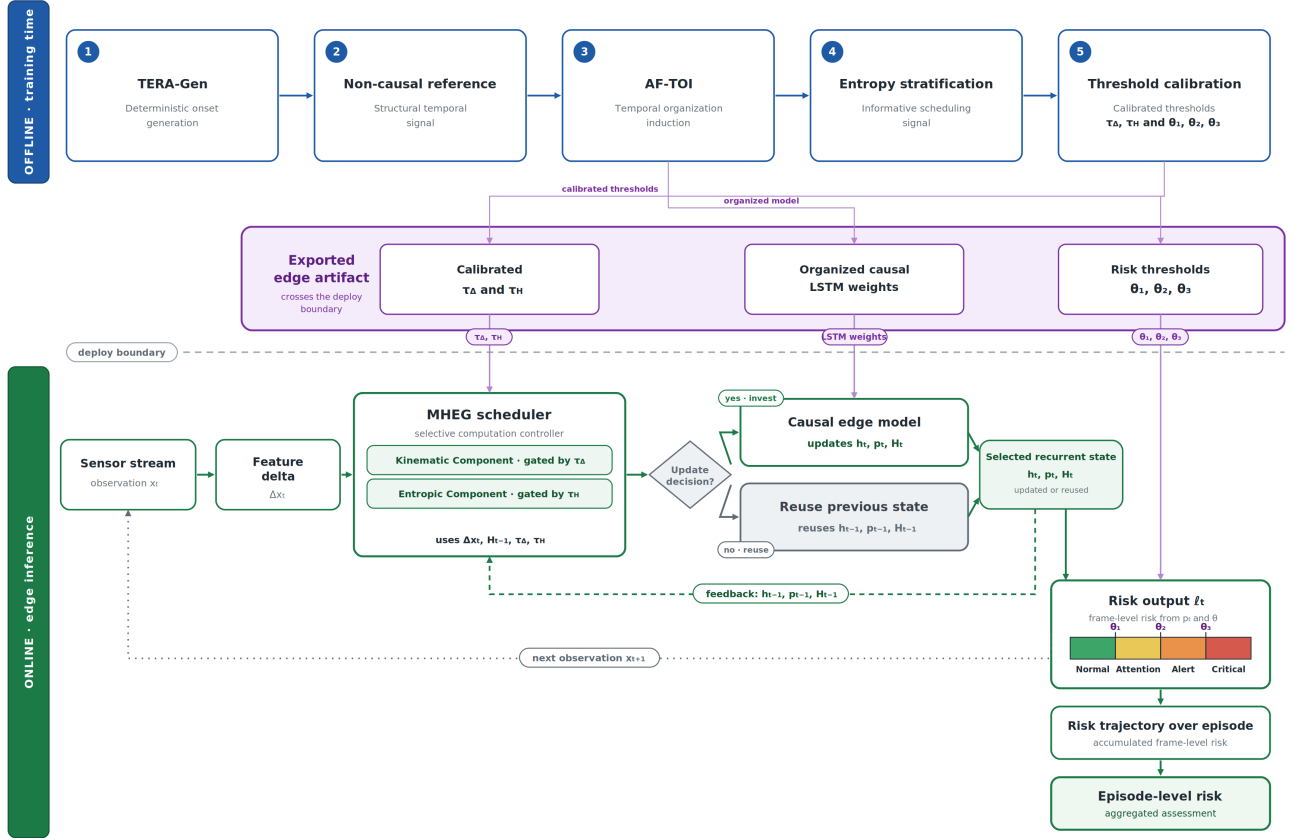


Figure 2: Operational architecture of TERA. Offline stages induce temporally organized trajectories and export the causal LSTM with calibrated thresholds. Online, at each timestep t , MHEG decides whether to update or reuse the recurrent state, from Δx_t , H_{t-1} , and the thresholds τ_Δ , τ_H , under coverage-preservation constraints.

predictive core is a two-layer unidirectional LSTM with per-step cost $\mathcal{O}(4(DH+H^2))$, compatible with automotive-grade edge budgets.

We evaluate the role of trajectory organization directly, using entropic separation, coverage stability, allocator selectivity, and TTD_{ef} across the factorial configurations. AF-TOI provides the organized condition, whereas the negative control removes it.

Figure 2 shows where each component enters the pipeline. The offline stages produce the causal model and the scheduling thresholds; the online stage uses them on the device for selective recurrent inference under the coverage constraint.

Operational problem formulation Table 2 lists the six objectives used in evaluation. Among them, TTD_{ef} is the metric that exposes the main trade-off. Traditional latency metrics such as TTD_{det} are computed only over detected episodes and ignore complete misses, which can make a system that misses a fraction of events report a low average latency on the rest. A scheduler may thus reduce apparent cost without changing the miss rate, and a coverage-blind metric would not notice. The effective detection delay integrates timeliness and coverage,

$$\text{TTD}_{\text{ef}} = \text{TTD}_{\text{det}} + \text{FR} \times T_{\text{ep}} \quad (1)$$

where TTD_{det} is the mean latency over detected episodes, FR is the missed-episode proportion, and T_{ep} is episode duration. This system-level form is the aggregate counterpart of the per-episode definition applied later in the protocol (Eq. 15), where a missed episode receives the full penalty T_{ep} . Under this metric, TERA is evaluated as a selective-computation mechanism for safety-critical vehicular edge settings, with operational formulation

$$\begin{aligned} \min \quad & \text{TTD}_{\text{ef}} \\ \text{subject to} \quad & \text{FR} \leq 0.05 \end{aligned} \quad (2)$$

Equation 2 is used only to state the evaluation criterion; no optimizer solves it during training or deployment. For configurations that satisfy $\text{FR} \leq 0.05$, suppression is considered useful only when it comes from skipping redundant regions rather than losing critical-event coverage.

The next subsections describe the three steps of this pipeline: onset anchoring, temporal-organization induction, and online selective computation.

3.1. Deterministic Temporal Anchoring (TERA-Gen)

Controlled induction and frame-level evaluation require a fixed onset reference. In naturalistic datasets, onset is fixed by

Table 2

Unified performance objective matrix of the TERA ecosystem. **Role:** **P** = primary; **S** = safety constraint; **A** = auxiliary evidence. **Meaning** summarizes the physical reading of each metric. [†] Proposed formulation; see Eq. 1.

Objective / Metric	Symbol	Role	Target	Meaning
Multimodal detection (Balanced F1)	F1	A	Max	Harmonic mean of precision and recall over the stream [46].
Predictive calibration (ECE)	ECE	A	Min	Gap between declared confidence and empirical frequency; improves as a byproduct of organization, not as an objective [10].
Partial timeliness (gross delay)	TTD _b	A	Min	Mean latency over detected episodes only; optimistic, as misses are excluded [47].
Effective timeliness	TTD _{ef}	P	Min	TTD _b + FR × T _{ep} (T _{ep} =10.0s); adds the miss penalty and arbitrates allocation. Proposed [†] .
Coverage stability (Failure Rate)	FR	S	≤ 0.05	Proportion of missed critical episodes; hard constraint on MHEG calibration [48].
Computational economy (suppression)	%Sup	A	Max	Fraction of frames whose LSTM update is suppressed within the FR constraint [6].

retrospective annotation with variability of ≈ 0.31 s at 9.6 fps, which masks real operational differences between schedulers by making it impossible to tell whether the allocator invested resources at emergence windows. TERA-Gen is a controlled laboratory for this, not a substitute for naturalistic data. By anchoring risk targets to SHRP2 odds ratios, TERA-Gen keeps the scenarios epidemiologically grounded while removing the annotation variability that would otherwise obscure scheduler behavior.

The generated scenarios include both progressive and abrupt risk transitions, enabling frame-precise quantification of TTD_{ef}, miss rate, and robustness. The system models 31 attributes across visual, kinematic, contextual, and acoustic domains grounded in traffic epidemiology.

Input vector structure Each $x_t \in \mathbb{R}^{31}$ contains four groups of variables: behavioral visual (seven driving classes, lateral and gaze deviation), kinematic (speed, longitudinal accelerations, gyroscopic signals), contextual (road type, precipitation, night driving, traffic density), and acoustic (siren, horn, phone ring, background noise). Two further indicators encode instantaneous detector reliability.

Epidemiological calibration Risk targets are projected onto [0.05, 1.00] by

$$risk_target \approx 0.05 + \frac{OR - 1}{12.5} \quad (3)$$

This mapping preserves ordinal severity; it is not intended as an estimate of absolute collision probability.

Table 3 gives the resulting values, and the continuous reference label combines mean trajectory and peak intensity,

$$y_{ep} = 0.7 \text{mean}(y_{frame}) + 0.3 p_{95}(y_{frame}) \quad (4)$$

Emergency regimes as design principle TERA-Gen builds critical episodes in two ramp patterns. The *progressive* regime (60% of critical scenarios) simulates gradual

Table 3

Risk behaviors calibrated by SHRP2 [13] and target parameterization.

Behavior	Risk Level	OR	<i>risk_target</i>
Normal driving	Normal	1.0	0.050
Hands-free conversation	Attention	1.3	0.074
Mild fatigue / drowsiness	Alert	4.0	0.290
Reaching for moving object	Critical	9.1	0.698
Manual cell phone dialing	Critical	12.2	0.946

development with a latent baseline of 8–20 timesteps and a ramp of ≈ 1.4 s. The *abrupt* regime (the remaining 40%) models fast-ramp perturbations < 0.6 s, a difficult case for coverage under adaptive computation. This regime tests whether the scheduler still allocates computation during a rapidly emerging risk.

Methodological choice of binary supervision Episodes use binary supervision (Normal vs. Critical). Section 4.7 shows that this onset-anchored supervision is enough to produce continuous ordinal probabilistic trajectories. Continuous SHRP2-calibrated targets generate temporally structured trajectories; training labels classify episodes as Normal or Critical, isolating temporal structuring from refined risk regression.

This fixed onset is what makes the alignment step meaningful. Without it, the causal trajectory could not be compared reliably to the structural reference around risk emergence. AF-TOI uses this anchor directly.

3.2. Temporal Organization Induction (AF-TOI)

AF-TOI is not used here as a conventional distillation procedure. Its purpose is to shape the causal model's temporal trajectories. It uses onset-oriented reference-to-target structural alignment during offline training, aiming at predictive dynamics that separate stable regions, pre-onset regions, and transition windows, which is distinct from model compression. Once that differentiation emerges, predictive

uncertainty stops behaving as a generic confidence estimate and becomes structured by the temporal phase of the episode.

The experiments below test whether this temporal structure is what makes entropy usable for scheduling. As trajectories acquire temporal structure, entropy stratifies across episode phases, separating routine intervals from critical transitions, and that stratification is the scheduling signal MHEG depends on. Resource allocation remains a downstream consequence of the induced structure, not a direct output of AF-TOI.

This induction step is needed because the deployed LSTM is causal: a unidirectional LSTM sees only accumulated history, so abrupt transitions with sub-0.6 s pre-onset windows risk weakly differentiated uncertainty and insufficient stratification.

During offline training, AF-TOI uses a non-causal BiLSTM as a reference. Because the reference has access to the full episode, it can represent the temporal shape of the transition. The causal target is then aligned to this structure, but only the causal model is deployed.

Onset-Oriented Temporal Loss Function The causal target is trained with the composite loss $\mathcal{L}_{\text{AF-TOI}}$, balancing supervised anchoring with alignment to the reference structural signal,

$$\mathcal{L}_{\text{AF-TOI}} = (1 - \lambda_{\text{TOI}})\mathcal{L}_{\text{hard}} + \lambda_{\text{TOI}}\mathcal{L}_{\text{soft}} \quad (5)$$

where $\lambda_{\text{TOI}} \in [0, 1]$ balances supervision and alignment. The supervised term is binary cross-entropy,

$$\mathcal{L}_{\text{hard}} = -[y_{\text{ep}} \log(\hat{p}_{\text{ep}}) + (1 - y_{\text{ep}}) \log(1 - \hat{p}_{\text{ep}})] \quad (6)$$

The structural term uses Kullback–Leibler (KL) divergence with temperature $T = 2.0$,

$$\mathcal{L}_{\text{soft}}(t) = T^2 \cdot \text{KL}\left(\sigma\left(\frac{z_r(t)}{T}\right) \parallel \sigma\left(\frac{z_s(t)}{T}\right)\right) \quad (7)$$

where $z_r(t)$ and $z_s(t)$ are the reference and causal-target logits at timestep t , respectively. The KL term penalizes discrepancies between the reference distribution and the causal target distribution, encouraging the target to reproduce the temporal structure learned by the reference.

Asymmetric temporal weighting $w(t)$ concentrates induction around onset regions,

$$w(t) = \begin{cases} w_{\text{pre}}, & \max(0, t_0 - k) \leq t < t_0 \\ \exp(-\alpha(t - t_0)), & t \geq t_0 \\ 1, & \text{otherwise} \end{cases} \quad (8)$$

This weighting depends on the deterministic onset supplied by TERA-Gen.

Three-Phase Structural Alignment Scheduler Because early coupling to the reference can destabilize gradients, $\lambda_{\text{TOI}}(e)$ is increased gradually,

$$\lambda_{\text{TOI}}(e) = \begin{cases} 0, & 1 \leq e \leq E_{\text{warm}} \\ \frac{e - E_{\text{warm}}}{E_{\text{ramp}} - E_{\text{warm}}}, & E_{\text{warm}} < e \leq E_{\text{ramp}} \\ 1.0, & E_{\text{ramp}} < e \leq E_{\text{total}} \end{cases} \quad (9)$$

with $E_{\text{warm}} = 10$, $E_{\text{ramp}} = 17$, $E_{\text{total}} = 80$ epochs. This schedule is part of AF-TOI rather than a tuning detail; removing it weakens coupling stability and the induced differentiation, as shown in the architectural ablation (Section 4.9).

The result is a stratified entropy signal: stable intervals tend to remain at low entropy, whereas pre-onset and transition windows show higher and more variable uncertainty. MHEG uses this signal at runtime.

3.3. Trustworthy Selective Computation (MHEG)

MHEG uses the stratification produced by AF-TOI; it is not designed to create that structure by itself. The gate is useful only when the trajectory already contains temporal structure. Its usefulness is therefore inherited from the offline-induced organization rather than from the gating rule alone. When the stratification is absent, the gate has no basis for discrimination, which is the failure reproduced by the negative control.

At runtime, MHEG decides at each timestep whether the embedded model runs a full LSTM update or reuses the previous recurrent state. The decision uses the AF-TOI-induced organization and rests on two components, synthesized in Figure 3.

Kinematic Component as Unconditional Resource Allocation Trigger

$$|\Delta x_t| = \max_{k \in \mathcal{K}} |x_t^{(k)} - x_{t-1}^{(k)}| \quad (10)$$

When this variation exceeds τ_{Δ} , the LSTM core runs regardless of entropic state. This trigger forces an update in abrupt episodes with sub-0.6 s pre-onset windows, where entropy may not rise before the transition begins.

Entropic Component as Trajectory-Dependent Scheduling Signal

$$H(p_{t-1}) = -p_{t-1} \log_2 p_{t-1} - (1 - p_{t-1}) \log_2 (1 - p_{t-1}) \quad (11)$$

When entropy exceeds τ_H , the scheduler invests full LSTM processing. In this setting, entropy is used as a scheduling variable, not as a direct error estimate. The scheduler does not need globally calibrated confidence; it needs uncertainty with temporal differentiation. The main scheduling failure considered here is not poor global calibration but absent temporal differentiation.

The discriminative power of the entropic component depends on trajectory organization. Without AF-TOI-induced

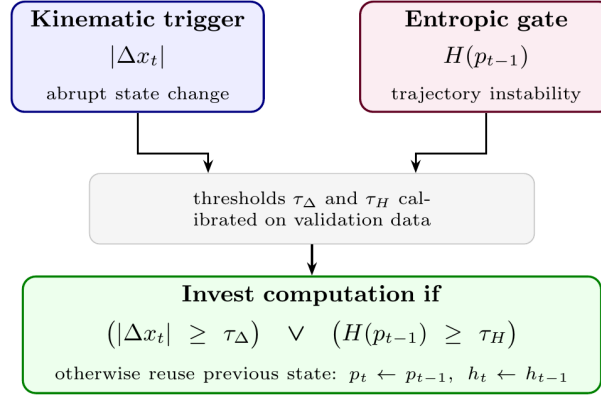


Figure 3: MHEG runtime computation scheduler structure. The kinematic component is an unconditional resource allocation trigger that forces LSTM computation whenever physical state changes abruptly, regardless of entropic state. The entropic component is the trajectory-dependent scheduling signal. When temporal organization is absent, entropy remains undifferentiated and the scheduler loses temporal discrimination power; when induced by AF-TOI, entropy stratifies near onset and the scheduler can allocate resources to emergence windows while safely suppressing redundant intervals.

differentiation, entropy sits in a narrow range and the scheduler loses selectivity; with it, phases stratify along the entropic axis and the criterion gains temporal discrimination. This is the dependency tested in Section 4.

Joint Threshold Calibration After freezing the causal target, a per-seed grid search over the validation set selects the threshold combination that maximizes suppression while keeping $FR \leq 0.05$. Calibration uses TERA-Gen synthetic data; online re-calibration is a necessary extension for production.

Scheduling Criterion via Logical Disjunction

$$\max_{k \in \mathcal{K}} |x_t^{(k)} - x_{t-1}^{(k)}| \geq \tau_\Delta \quad \vee \quad H(p_{t-1}) \geq \tau_H \quad (12)$$

When neither condition holds, the LSTM update is suppressed and previous values propagate. This rule reflects the safety asymmetry of the problem: an extra update is cheaper than skipping computation during a transition window.

Operational Regimes The thresholds define three operating regimes. As thresholds $\rightarrow 0$, the system runs full continuous inference: nothing is suppressed, cost is maximal, FR is zero. As thresholds $\rightarrow \infty$, suppression is total and equivalent to halting all LSTM updates: cost is zero, FR is maximal. Between these, per-seed calibration on validation data yields the trustworthy hybrid regime the ecosystem occupies, maximizing economy under $FR \leq 0.05$. The online execution procedure is given next.

3.4. Runtime Inference and Risk Assessment

Algorithm 1 gives the runtime procedure executed on the edge device. The continuous probability p_t is mapped to four discrete risk levels.

Algorithm 1 Efficient inference cycle of the MHEG runtime computation scheduler

Require: Weights θ_{LSTM} ; thresholds $\tau_\Delta, \tau_H, \theta_1=0.15, \theta_2=0.38, \theta_3=0.50; h_0=0; p_0=0$

Ensure: Level $\ell_t \in \{0, 1, 2, 3\}$ emitted at each timestep t

```

1: for each frame  $x_t$  do
2:    $\Delta x_t \leftarrow \max_{k \in \mathcal{K}} |x_t^{(k)} - x_{t-1}^{(k)}|$ 
3:    $H_t \leftarrow H(p_{t-1})$ 
4:   if  $(\Delta x_t \geq \tau_\Delta)$  or  $(H_t \geq \tau_H)$  then
5:      $(p_t, h_t) \leftarrow \text{LSTM}(x_t, h_{t-1}; \theta_{\text{LSTM}})$  [Invest computation]
6:   else
7:      $p_t \leftarrow p_{t-1}; h_t \leftarrow h_{t-1}$  [Reuse buffer – suppress computation]
8:   end if
9:    $\ell_t \leftarrow \mathbf{1}[p_t \geq \theta_3] \cdot 3 + \mathbf{1}[\theta_2 \leq p_t < \theta_3] \cdot 2 + \mathbf{1}[\theta_1 \leq p_t < \theta_2] \cdot 1$ 
10:  Emit  $\ell_t$ 
11: end for

```

$$\ell_t = \begin{cases} 0 & \text{Normal, } p_t < 0.15 \\ 1 & \text{Attention, } 0.15 \leq p_t < 0.38 \\ 2 & \text{Alert, } 0.38 \leq p_t < 0.50 \\ 3 & \text{Critical, } p_t \geq 0.50 \end{cases} \quad (13)$$

3.5. Computational Complexity and Edge Deployment

The final step is to verify that only lightweight components remain active at runtime.

Table 4 separates the components used only during training from those that remain active on the device. The highest-cost modules (TERA-Gen, the non-causal BiLSTM

Table 4

Computational complexity profile of TERA ecosystem components.

Component	Functional role	Complexity	Executes at	On device
TERA-Gen	Deterministic onset episode generation	$\mathcal{O}(E \cdot T \cdot D)$	Training only	✗
Non-causal BiLSTM reference arch.	Source of temporal structural signal for AF-TOI alignment	$\mathcal{O}(8(DH+H^2))/\text{step}$	Training only	✗
AF-TOI	Temporal organization induction in causal target architecture	$\mathcal{O}(T)$ per episode	Training only	✗
MHEG calibration	Per-seed grid search of scheduler thresholds	$\mathcal{O}(L \cdot E_{\text{val}} \cdot T)$	Training only	✗
Causal LSTM (embedded target arch.)	On-device recurrent inference	$\mathcal{O}(4(DH+H^2))/\text{step}$	Continuous runtime	✓
MHEG (runtime)	Runtime computation scheduler / adaptive inference controller	$\mathcal{O}(1)/\text{step}$	Continuous runtime	✓
Hidden state h_t	Reused recurrent memory (buffer reuse on suppression)	$\mathcal{O}(H)$	Continuous runtime	✓
Thresholds $(\tau_\Delta, \tau_H, \theta)$	Constant scalar scheduling parameters	$\mathcal{O}(1)$	Continuous runtime	✓

Legend: E : training episodes; T : episode temporal length; D : input dimensionality; H : LSTM hidden dimension; L : number of combinations evaluated in grid search.

reference, AF-TOI, and MHEG calibration) stay offline. At runtime only the causal LSTM target, the MHEG scheduler, the recurrent state, and the calibrated thresholds are active on device.

The hidden and cell-state buffers require 1,024 bytes of static RAM. The causal LSTM with two layers and 64 hidden units has ≈ 58 K parameters (≈ 228 KB in FP32, ≈ 57 KB in INT8), which is within the memory range of platforms such as STM32H7, Cortex-M55 with Helium, Jetson Nano, and Qualcomm SA8155P.

4. Experimental Evaluation

The evaluation is organized into six movements, each addressing a specific part of the hypothesis. Movements 1 and 2 examine the mechanism: whether AF-TOI induces temporal organization in the causal target and whether that organization confers scheduling selectivity to MHEG.

Movement 3 quantifies the operational consequence on coverage, TTD_{ef} , and efficiency. The last three probe the nature of the effect: whether it comes from a reorganized failure distribution, whether it is stable across seeds and regimes, and whether it extends to intermediate risk.

Architectural ablation and embedded viability close the section.

4.1. Experimental Protocol

The protocol is designed to make temporal organization observable at frame level. This allows the link between organization and selective computation to be tested against a deterministic onset instead of inferred from aggregate metrics.

Synthetic multimodal dataset TERA-Gen synthesizes episodes with 13 training recipes, N01–N03 for normal driving and C01–C10 for critical risk. Six exclusive recipes form the borderline set, A01–A03 for the Attention level and L01–L03 for Alert.

Table 5

Recipes used in TERA-Gen synthetic episode generation.

Set	Level	Recipes	Application
Training	Normal	N01–N03	Train, validation, test
	Critical	C01–C10	Train, validation, test
Borderline	Attention	A01–A03	Evaluation only
	Alert	L01–L03	Evaluation only

Training recipes were generated for the main experimental pipeline; borderline recipes were generated independently and never exposed during training, serving exclusively as a held-out evaluation set for Movement 6 (Section 4.7).

Table 5 summarizes the full recipe taxonomy.

The dataset consolidates 4,160 episodes across four seeds (42–45) with 70/15/15% split. Each sequence has $T = 96$ timesteps at ≈ 9.6 fps, $D = 31$, and deterministic onset t_0 at frame-level precision.

That onset makes the induced organization verifiable, since it fixes the reference against which entropic differentiation and scheduler selectivity are measured.

Table 6 details the full parameterization for reproducibility.

Threshold configuration and calibration Thresholds τ_Δ and τ_H are calibrated per seed via grid search on the validation partition, selecting the configuration that maximizes suppression under $\text{FR} \leq 0.05$.

A detection is valid only when $p_t \geq \theta_{\text{TDD}}$ for $m = 3$ consecutive timesteps.

$$\text{TTD}_{\text{det}} = (t_{\text{det}} - t_0) \cdot \Delta t, \quad t_{\text{det}} = \min\{t \geq t_0 \mid p_t, p_{t+1}, p_{t+2} \geq \theta_{\text{TDD}}\} \quad (14)$$

Table 6

Architecture, optimization, and temporal organization induction hyperparameters.

Component	Symbol / Variable	Value	Role in Pipeline
General Architecture and Data	D (input_dim)	31	Multimodal vector x_t dimensionality
	H (hidden_dim)	64	Hidden units per recurrent layer
	num_layers	2	Stacked recurrent layers
	dropout	0.1	Structural regularization factor
Non-Causal BiLSTM Reference Arch. (<i>Offline</i>)	bi	True	Enables non-causal bidirectional processing
	factor	2	Hidden state multiplier ($2 \times H = 128$)
	lr	0.001	Adam optimizer learning rate
	epochs	60	Training epochs with combined $\mathcal{L}_{\text{ref}}$ loss
Loss Function $\mathcal{L}_{\text{AF-TOI}}$ (Causal Target Arch.)	α (onset_exp_alpha)	0.15	Post- t_0 decay rate (Eq. 8)
	T (temp_fixed)	2.0	KL div. smoothing temperature (Eq. 7)
	pre_onset_push_weight	3.0	Amplified weight at latent window
	hidden_align_window	12	Pre-onset window extent (frames)
	λ_{late} (lambda_late)	3.5	Late Penalty multiplier
	θ_{late} (theta_late)	0.15	Causal target reactivity threshold
Structural Alignment Scheduler	E_{warm} (warmup_epochs)	10	Initial epochs with pure hard supervision
	E_{ramp} (rampup_epochs)	17	Linear growth window of $\lambda_{\text{TOI}}(e)$
	E_{total} (epochs)	80	Total causal target optimization epochs
	$[\lambda_{\text{start}}, \lambda_{\text{end}}]$	[0.2, 1.0]	Soft alignment loss coupling bounds

$$\text{TTD}_{\text{ef}} = \begin{cases} \text{TTD}_{\text{det}}, & \text{if } t_{\text{det}} \text{ exists} \\ T_{\text{ep}} = 10.0 \text{ s}, & \text{otherwise (missed episode)} \end{cases} \quad (15)$$

Infrastructure and significance tests Latency is measured in unit-batch mode ($N_{\text{batch}} = 1$) on an NVIDIA RTX 3050 Ti GPU with PyTorch 2.0 and 100-iteration warmup.

Hypotheses use the paired Wilcoxon test [49], with effect quantified by Cliff's delta [50] and 95% confidence intervals by recipe-stratified bootstrap [51].

4.2. Movement 1. Temporal Mechanism

Movement 1 examines whether AF-TOI changes the causal target's trajectory structure at frame level.

This mechanistic evidence precedes any attribution of allocation gains to organization.

In the Baseline, H_t stays poorly differentiated across driving phases: phase distributions collapse into a narrow homogeneous range with no visible separation between stable and transition zones (Figure 4(a)). The scheduling signal is then indiscriminating at the periods of highest criticality, with no temporal basis to distinguish emergence windows from redundant intervals.

With AF-TOI, phases stratify along the entropic axis (Figure 4(b)): Normal anchors near zero, Pre-onset converges to the central region ($\frac{\ln 2}{2}$), and Post-onset accumulates at the maximum ($\ln 2$). This is the pattern that gives MHEG a usable scheduling signal.

As a byproduct of the induction, alignment between declared confidence and empirical frame-level accuracy improves, with $\text{ECE}_{\text{frame}} = 0.044$ for AF-TOI versus 0.129

for Baseline (Figure 4(c)). This calibration gain accompanies stratification but is not the objective.

Figure 5 adds the episode-level view. In AF-TOI+MHEG, suppressed intervals concentrate in low-entropy periods while LSTM computation is preserved near onset, so the scheduler invests at emergence windows and reuses buffers at redundant timesteps. In the Baseline, suppressions distribute uniformly and the scheduler allocates without temporal basis.

With AF-TOI, changes in H_t become more aligned with risk transitions, making entropy usable by the scheduler. Movement 2 tests whether this translates into scheduler selectivity.

4.3. Movement 2. Scheduler Selectivity

The next question is whether this reorganization changes the scheduler's allocation pattern.

Figure 5 already shows the resulting asymmetry: with AF-TOI, suppressed intervals concentrate in pre-onset periods while the onset neighborhood is consistently preserved, so computation goes to emergence windows in both regimes.

In the Baseline, suppression is uniform and the criterion has no temporal semantics.

Figure 6 extends this over 120 critical sequences from seed 42. In the Baseline, H_t decreases after t_0 , giving no discrimination signal. With AF-TOI, H_t rises near onset, shifting MHEG's investment to emergence timesteps. The alignment between entropic elevation and onset proximity is how the scheduler acquires temporal discrimination.

The pattern is therefore conditional on trajectory organization. When the structure is present, MHEG concentrates computation near emergence timesteps; when it is absent, the gate lacks a temporal criterion.

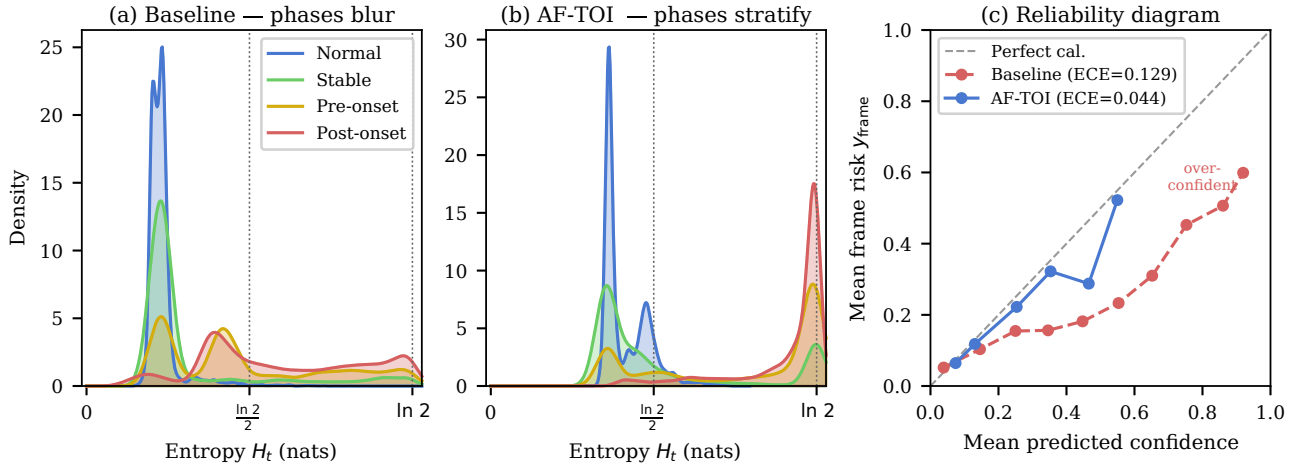


Figure 4: Trajectory organization across driving phases for seed 42. Panels compare Baseline entropy structure, AF-TOI-induced phase stratification, and frame-level calibration alignment ($\text{ECE}_{\text{frame}} = 0.044$ vs. 0.129 for Baseline).

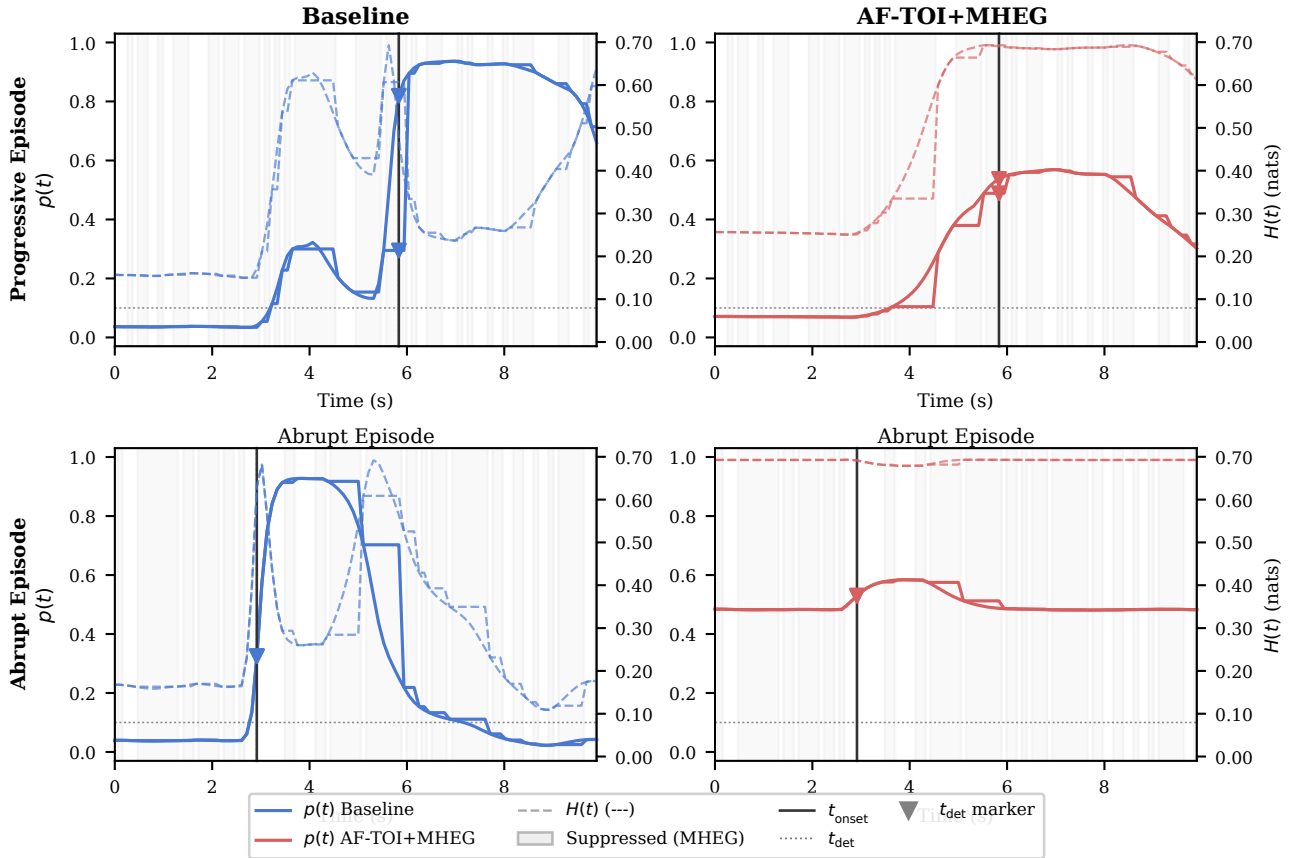


Figure 5: Frame-by-frame temporal evidence for seed 42. Columns show the Baseline and AF-TOI+MHEG configurations; rows show progressive and abrupt episodes. Solid curves denote p_t , dashed curves denote H_t , vertical lines mark t_{onset} , markers ∇ indicate t_{det} , and gray regions indicate MHEG buffer reuse.

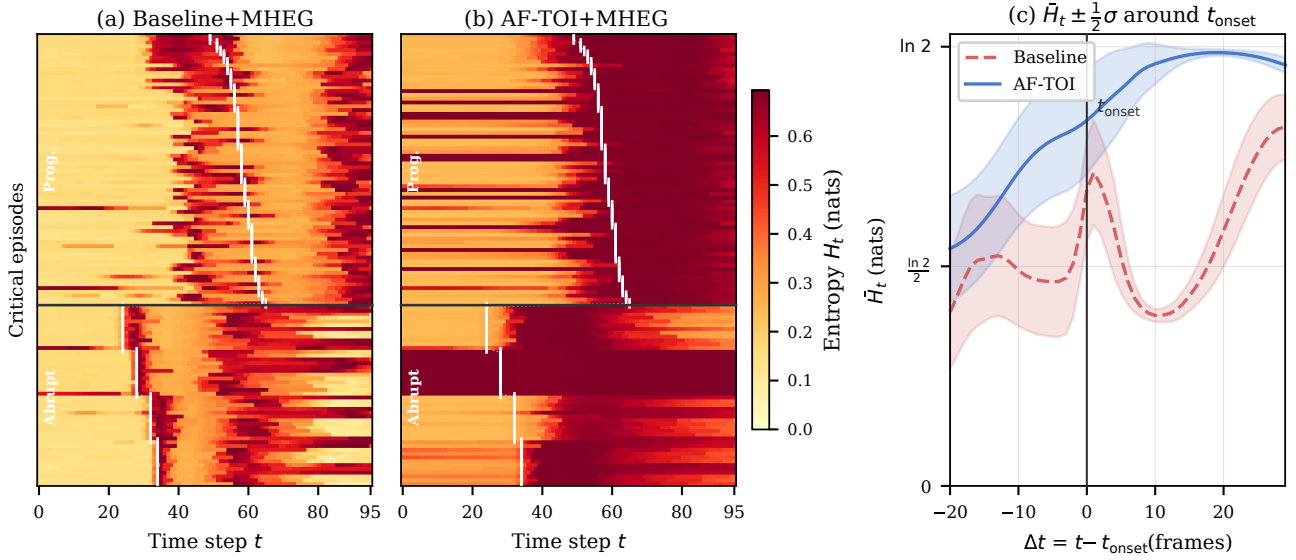


Figure 6: Temporal H_t profile across 120 critical episodes for seed 42. The heatmaps compare Baseline and AF-TOI entropy trajectories around the deterministic onset reference.

Table 7
2x2 factorial design and per-configuration seed coverage.

Configuration	AF-TOI	MHEG	Seeds	Methodological Role
Baseline Fixed	✗	✗	4/4	Basal reference without temporal organization
Baseline+MHEG	✗	✓	4/4	Negative control: unsafe scheduling without temporal org.
AF-TOI Fixed	✓	✗	4/4	Temporal organization isolated; no scheduler
AF-TOI+MHEG	✓	✓	4/4	Complete ecosystem: trustworthy selective computation

Seeds: number of seeds (42–45) producing valid runs per configuration.

4.4. Movement 3. Operational Consequence

The 2x2 factorial study evaluates how temporal organization and selective computation affect coverage, delay, and cost.

The negative control is the most informative point, measuring what happens to allocation when selective computation is applied to unorganized trajectories.

Table 7 details the design; global results are in Table 8.

The negative control as unsafe adaptive computation
Applying MHEG to unorganized trajectories has non-neutral effects on allocation quality. Baseline+MHEG holds $FR = 0.040 \pm 0.052$ at the fixed-baseline level, so it does not

reduce misses, and it still degrades TTD_{ef} by 11.8%, from 0.602 s to 0.673 s. The scheduler allocates without temporal discrimination, suppressing emerging-risk timesteps as often as redundant ones. This is the unsafe regime: apparent per-frame cost drops with no coverage guarantee while the effective delay worsens.

This indicates that the failure comes from applying the gate to an unorganized trajectory, rather than from the gating rule alone.

The complete ecosystem and trustworthy selective computation
Under AF-TOI organization, MHEG operates on differentiated trajectories. The miss rate is preserved at 0.004 ± 0.008 , with TTD_{ef} rising only 0.015 s relative to AF-TOI fixed. The latency impact of scheduling is about 4.7x smaller in the complete ecosystem than in the negative control (0.015 s vs. 0.071 s), and absolute TTD_{ef} with active scheduling is 0.101 s versus 0.673 s.

In this synthetic setting, the results indicate that efficiency and coverage can coexist when the scheduler reads temporally organized trajectories.

Regime asymmetry in progressive episodes
AF-TOI Fixed shows $TTD_p = 1.419 \pm 0.391$ s in progressive episodes, above the Baseline (0.476 ± 0.099 s): reorganizing dynamics at transition regions can delay classification consolidation in slowly developing episodes. The benefit concentrates in the abrupt regime, where the miss rate falls from 0.099 to 0.010, exactly where misses cost most.

TTD_b vs. TTD_{ef}
Gross delay counts only detected episodes and reads optimistically because it hides misses. TTD_{ef} corrects this for edge systems where coverage across all episodes matters more than mean latency on successful detections.

Table 8Global factorial study results under continuous causal streaming (mean $\pm$ inter-seed standard deviation).

Model	MHEG	F1 $\uparrow$	ECE $\downarrow$	FR $\downarrow$	TTD _p (s) $\downarrow$	TTD _b (s) $\downarrow$	TTD _{ef} (s) $\downarrow$	ms/f $\downarrow$	%Sup $\uparrow$
LSTM-Baseline	No	0.357 $\pm$ 0.096	0.690 $\pm$ 0.007	0.040 $\pm$ 0.052	0.476 $\pm$ 0.099	0.206 $\pm$ 0.099	0.602 $\pm$ 0.608	1.015 $\pm$ 0.130	0.000 $\pm$ 0.000
LSTM-Baseline	Yes	0.354 $\pm$ 0.101	0.690 $\pm$ 0.008	0.040 $\pm$ 0.052	0.370 $\pm$ 0.150	—	0.673 $\pm$ 0.682	0.400 $\pm$ 0.148	61.457 $\pm$ 8.819
LSTM-AF-TOI	No	0.751 $\pm$ 0.275	0.497 $\pm$ 0.221	0.004 $\pm$ 0.008	1.419 $\pm$ 0.391	0.044 $\pm$ 0.052	0.086 $\pm$ 0.136	0.920 $\pm$ 0.062	0.000 $\pm$ 0.000
LSTM-AF-TOI	Yes	0.751 $\pm$ 0.275	0.497 $\pm$ 0.221	0.004 $\pm$ 0.008	0.970 $\pm$ 0.266	0.060 $\pm$ 0.071	0.101 $\pm$ 0.155	0.473 $\pm$ 0.079	48.657 $\pm$ 7.335

TTD_p: latency in progressive regime. TTD_b: gross latency over detected episodes (partial, optimistic). TTD_{ef}: effective detection delay with miss penalty (primary arbiter). ECE: auxiliary indicator. %Sup: fraction of frames where MHEG suppressed computation (buffer reuse).

Table 9

Cliff's delta and paired Wilcoxon test: Baseline Fixed vs. AF-TOI Fixed.

Metric	$\delta_{\text{Cliff}} \uparrow$	Magnitude	p-value
ECE	+1.000	Large	0.0625
F1	+1.000	Large	0.0625

$n = 4$. p -floor reflects rank saturation, not absent effect. $\delta_{\text{Cliff}} = +1.000$: dominance across all seeds. Magnitudes per Romano et al. [52].

Statistical analysis The AF-TOI F1-Score (0.751 ± 0.275 , $\text{CV} \approx 37\%$) shows greater inter-seed variance than Baseline (0.357 ± 0.096), reflecting sensitivity of the alignment process to initialization. With $n = 4$ seeds the Wilcoxon test saturates ranks (p -floor 0.0625), so analysis relies on $\delta_{\text{Cliff}} = +1.000$ and direct operational means (FR 0.004 vs. 0.040; TTD_{ef} 0.086 s vs. 0.602 s). Table 9 summarizes.

Table 10 stratifies by operational regime, confirming that the complete ecosystem preserves coverage in both progressive and abrupt profiles.

Figure 7 synthesizes the trade-off. Baseline Fixed and Baseline+MHEG occupy identical FR positions, so selective computation without temporal organization gives no coverage benefit; AF-TOI+MHEG is the only configuration reaching low cost and FR below the safety threshold at once.

4.5. Movement 4. Distributional Structure

Mean TTD_{ef} can mask heterogeneous per-episode behavior.

Figures 8 and 9 show the pattern: AF-TOI histograms concentrate near zero, while variants without organization spread widely, with 19 episodes stuck at the 10 s miss penalty, precisely the episodes where the scheduler failed to invest at emergence windows.

The high standard deviations of the non-AF-TOI configurations are not stochastic noise but the 19 severely penalized episodes where computation was suppressed at critical windows.

With AF-TOI, the miss population collapses from 19 to 2, so the induction restructures the failure distribution rather than merely lowering mean latency.

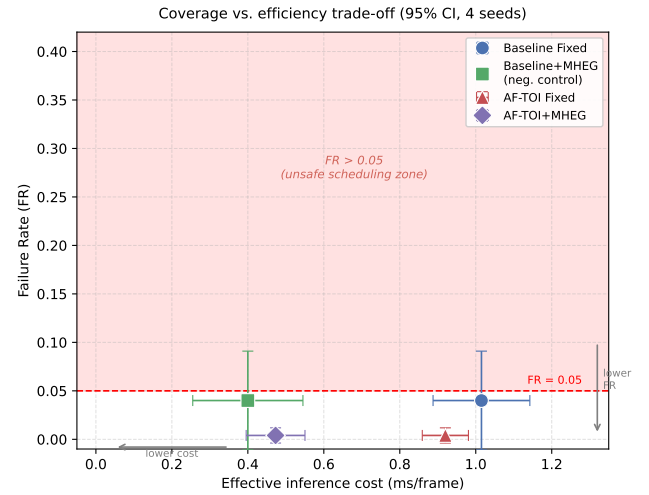


Figure 7: Global miss rate versus effective inference cost per frame (CI 95%, seeds 42–45). The dashed line marks the FR = 0.05 safety boundary; the shaded area denotes the unsafe scheduling region.

Baseline Fixed and Baseline+MHEG share identical miss counts, confirming no distributional coverage benefit from selective computation without organization.

Adding MHEG to already organized trajectories does not shift the profile, since the scheduler suppresses genuinely redundant intervals without creating new failures.

4.6. Movement 5. Robustness

Figure 10 shows performance across four seeds with 95% CI. AF-TOI+MHEG stays below FR = 0.05 in every run with lower variability than the other configurations, while variants without organization cross the coverage threshold in at least one seed.

The robustness results therefore reinforce the same point: selective computation is safe only when the temporal structure is consistently present.

Table 11 shows α sensitivity. The value $\alpha = 0.15$ gives the best balance, with stable F1 while FR and ECE grow more sensitive outside this region, consistent with organization quality depending more on gradient distribution near onset than on global discriminative capacity.

Table 10Performance stratified by operational regime (mean $\pm$ sd inter-seed, seeds 42–45).

Configuration	Progressive Regime (60%)		Abrupt Regime (40%)		Global	
	FR	TTD _p (s)	FR	TTD _b (s)	FR	TTD _{ef} (s)
Baseline Fixed	0.000 $\pm$ 0.000	0.476 $\pm$ 0.099	0.099 $\pm$ 0.131	0.544 $\pm$ 0.308	0.040 $\pm$ 0.052	0.602 $\pm$ 0.608
Baseline + MHEG	0.000 $\pm$ 0.000	0.370 $\pm$ 0.150	0.099 $\pm$ 0.131	0.573 $\pm$ 0.352	0.040 $\pm$ 0.052	0.673 $\pm$ 0.682
AF-TOI Fixed	0.000 $\pm$ 0.000	1.419 $\pm$ 0.391	0.010 $\pm$ 0.021	0.046 $\pm$ 0.084	0.004 $\pm$ 0.008	0.086 $\pm$ 0.136
AF-TOI + MHEG	0.000 $\pm$ 0.000	0.970 $\pm$ 0.266	0.010 $\pm$ 0.021	0.063 $\pm$ 0.117	0.004 $\pm$ 0.008	0.101 $\pm$ 0.155

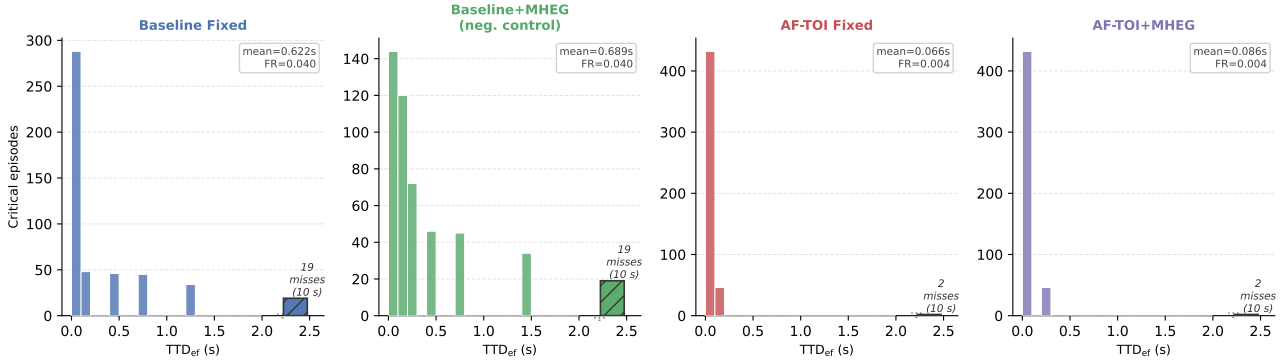
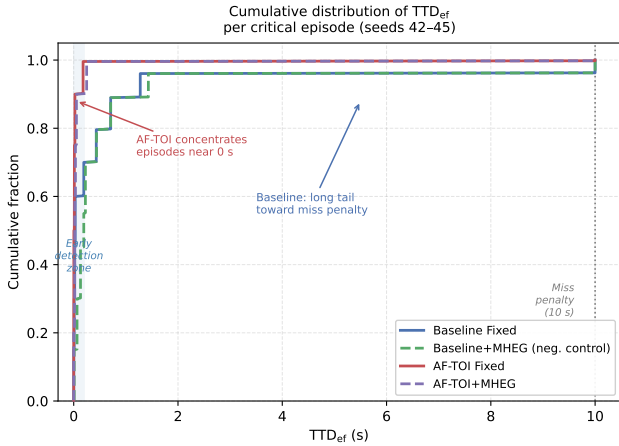
Distribution of TTD_{ef} per critical episode (seeds 42–45)**Figure 8:** TTD_{ef} frequency distributions per critical episode across seeds 42–45 ($n = 480$ per configuration). The 10 s value denotes the miss penalty used in TTD_{ef}.**Figure 9:** Cumulative distribution functions of TTD_{ef} per critical episode across seeds 42–45 ($n = 480$ per configuration).

Table 12 checks robustness to the miss penalty. The relative ordering across configurations holds for 5, 10, and 15 s penalties, so the complete-ecosystem advantage does not depend on the specific T_{ep} .

4.7. Movement 6. Emergent Organization

Movement 6 examines a property not directly optimized: whether AF-TOI organization extends to intermediate risk, producing outputs coherent with SHRP2 gradients.

Table 11Sensitivity to parameter α (AF-TOI without MHEG, seed 42).

α	F1 $\uparrow$	FR $\downarrow$	ECE $\downarrow$	TTD _p (s) $\downarrow$
0.05	≈ 0.88	≈ 0.05	≈ 0.53	≈ 0.40
0.15	0.876	≈ 0	0.324	1.155
0.30	≈ 0.90	≈ 0.008	≈ 0.54	≈ 0.60
0.50	≈ 0.89	≈ 0.010	≈ 0.56	≈ 0.55

In borderline scenarios unseen during training, the Attention regime records $\bar{p}_{\max} = 0.048 \pm 0.195$ (5.8% activations above θ) and Alert $\bar{p}_{\max} = 0.380 \pm 0.415$ (48.8%), evaluated on seed 42.

Table 13 and Figure 11 summarize.

The results give preliminary evidence that the induction extends beyond the training recipes. Normal and Attention episodes stay mostly below the activation threshold, while Alert and Critical trigger activation with progressively higher frequency, qualitatively aligned with increasing risk exposure despite no explicit severity objective during training.

The Alert–Critical mean ordering is not monotonic (0.380 vs. 0.271), so average probability should not be read as an ordinal severity measure.

The large variance in both groups and the heterogeneous recipes suggest that, in this controlled setting, activation frequency was a more stable summary of the system's response to increasingly hazardous situations.

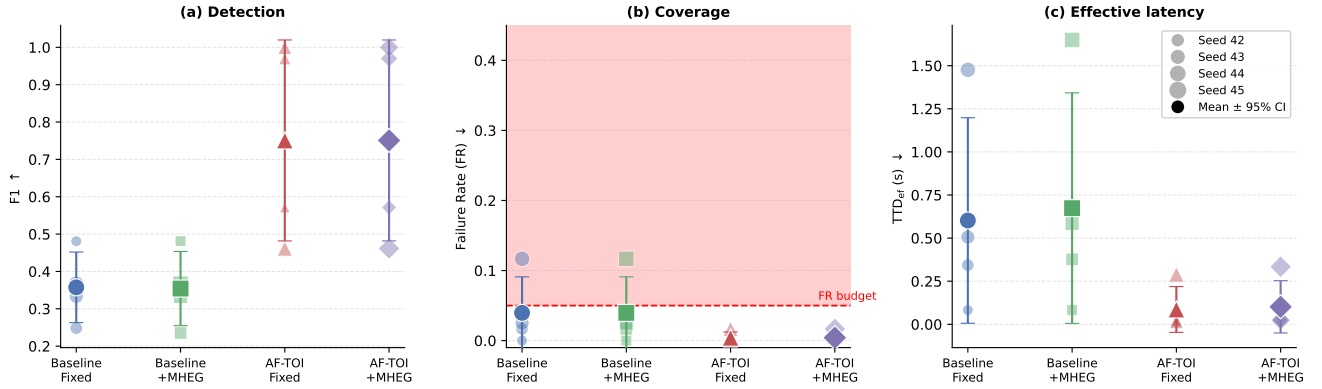
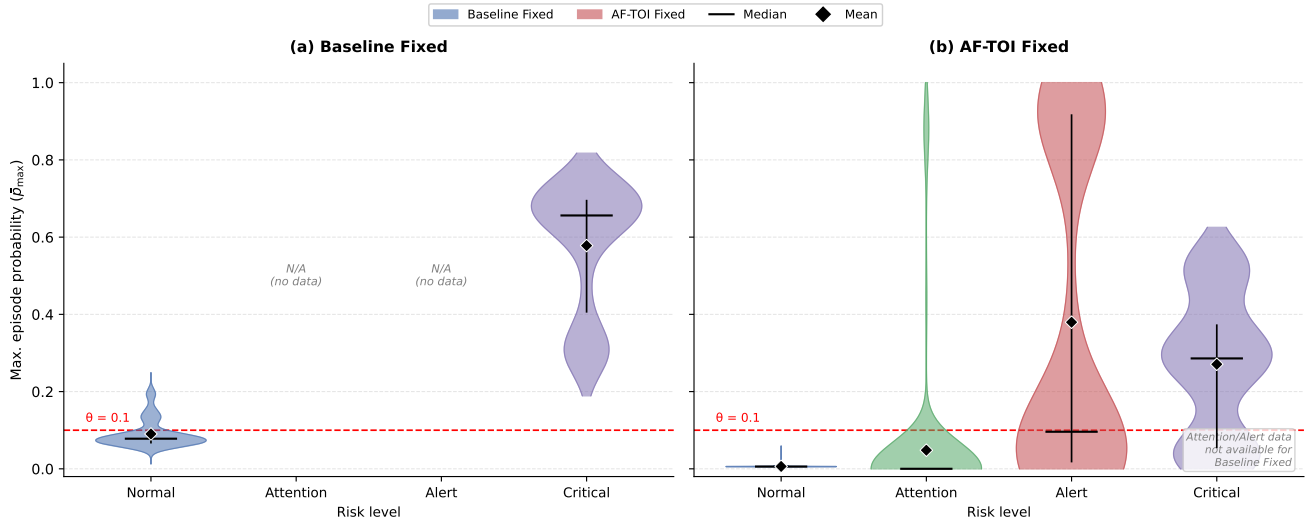
Per-seed performance across four independent partitions
(95% CI, seeds 42-45)

Figure 10: Per-seed performance across four independent partitions (CI 95%). The dashed line marks the FR = 0.05 safety boundary.

Table 12

 TTD_{ef} sensitivity as a function of miss penalty.

Configuration	Penalty 5.0 s	Penalty 10.0 s	Penalty 15.0 s
Baseline Fixed	0.406	0.606	0.806
Baseline+MHEG	0.478	0.678	0.878
AF-TOI Fixed	0.064	0.084	0.104
AF-TOI+MHEG	0.080	0.100	0.120

 Computed analytically from $TTD_{ef} = TTD_b + FR \times T_{ep}$.

Figure 11: $\bar{p}_{max}$ distributions by severity level in borderline episodes for seed 42. Baseline Fixed contains Normal and Critical levels; AF-TOI Fixed includes Normal, Attention, Alert, and Critical levels with activation fractions reported in the plot.

The evidence therefore does not support the claim that AF-TOI learned a severity scale. It suggests instead that the induction produces decision dynamics that grow progressively more likely to activate as criticality increases, even for intermediate-risk conditions unseen in training.

Whether this holds on naturalistic data, and whether full ordinal separation can emerge under broader conditions, remain open (Section 5).

Table 13

Probabilistic response in borderline episodes (seed 42, AF-TOI Fixed configuration).

Risk Level	Behavior	$\bar{p}_{\max}$	Fraction
Normal	Stable baseline driving	0.006 ± 0.001	0.0%
Attention	Environmental perturbation	0.048 ± 0.195	5.8%
Alert	Mild distraction at low speed	0.380 ± 0.415	48.8%
Critical	Severe distraction with imminent risk	0.271 ± 0.178	70.8%

Fraction: proportion of episodes with $\bar{p}_{\max} \geq \theta = 0.10$. Critical episodes from the main test set; Attention and Alert from independently generated borderline recipes.

4.8. Epistemological Control via Post-hoc Calibration

A complementary question is whether temporal organization could be approximated by post-hoc calibration of Baseline probabilities.

Three techniques were applied: temperature scaling (TS) [10], Platt scaling (PS) [39], and isotonic calibration (IC) [40].

MHEG was applied to each variant with thresholds recalibrated per seed to maximize suppression under $FR \leq 0.05$, yielding Table 14.

Probabilistic alignment improves in all variants. Uncalibrated Baseline ECE is ≈ 0.035 ; after isotonic calibration it drops to 0.0075 (reduction 0.0274). Thus, the calibration procedures improve probabilistic alignment as expected.

Their operational profiles under MHEG differ sharply. Temperature scaling degrades TTD_{ef} from 0.2324 s to 0.4377 s, reproducing the unsafe regime of the negative control. Platt scaling and isotonic calibration reach lower TTD_{ef} under MHEG (0.1221 s and 0.0527 s), but this improvement should not be interpreted as temporal organization.

The key property is that post-hoc calibration is a monotonic transformation of output probabilities: it preserves the ordering of $p(t)$ along time and cannot reorganize the temporal structure of $H(t)$.

Figure 12 confirms this. In panel (a), TS, PS, and IC preserve the Baseline temporal morphology, and the entropy profile around onset stays undifferentiated. The H_{strat} scores match the Baseline across seeds (0.7364 ± 0.1082 for TS, 0.7694 ± 0.0573 for IC), so no technique reorganizes temporal entropy dynamics. In panel (b), AF-TOI structurally shifts the trajectory, elevating $H(t)$ near onset and suppressing it in stable phases.

The lower TTD_{ef} for PS and IC under MHEG does not come from temporal reorganization. Both redistribute the probability scale, and IC in particular compresses probabilities toward empirical frequencies, shifting the operating regime of the entropy-based gate without altering the temporal ordering of uncertainty.

When MHEG thresholds are re-calibrated to that scale, lower suppression cost is achievable at a given coverage. This is a threshold calibration effect, not an organizational one: H_{strat} stays within the Baseline range and the operational

frontier in Figure 13 does not move to a qualitatively distinct location.

AF-TOI, by contrast, induces an organization regime not reproduced by probability-scale adjustment, with H_{strat} values that systematically exceed the calibrated Baseline range across seeds.

Figure 13 shows the ECE- TTD_{ef} operational locus. Temperature scaling overlaps the Baseline frontier. Platt scaling and isotonic calibration reach lower TTD_{ef} through threshold recalibration but stay on a frontier structurally distinct from AF-TOI, which reduces FR and TTD_{ef} simultaneously in a way post-hoc calibration does not reproduce.

Figure 14 reinforces this in the pre-onset domain: the pre-onset stratification score stays within the Baseline range across all three techniques and all seeds, while AF-TOI raises it consistently. Probability-scale correction does not modify the internal representation or how it responds to risk proximity; it rescales declared confidence at the output without inducing the temporal differentiation reliable scheduling needs.

This distinction is important for interpreting the control experiment. Reaching low TTD_{ef} through threshold recalibration to a redistributed probability scale differs from reaching it through induced temporal differentiation. In the former, the gain is contingent on the calibrator's particular distribution and may not survive distributional shift, since the gate lacks semantic grounding in the transition dynamics. In the latter, the scheduler exploits a structural trajectory property deliberately induced during training and tied to the causal dynamics of the process. The H_{strat} evidence and the frontier in Figure 13 show the two regimes are empirically distinguishable, with AF-TOI occupying the organizationally grounded one.

4.9. Architectural Ablation of the Non-Causal Reference

To test whether the organization gains depend on the specific BiLSTM topology, an ablation used a simplified protocol (60 epochs, no three-phase scheduler).

Two deliberate simplifications explain why the absolute values in Table 15 differ from the main study and must not be compared directly.

First, the three-phase alignment scheduler, which stabilizes AF-TOI coupling, was omitted.

Second, MHEG was excluded to isolate reference topology, which removes the coverage-preserving layer and drives FR to deployment-unacceptable levels.

The question is strictly comparative: which reference topology produces greater relative organization in the causal target under otherwise identical conditions.

With the causal LSTM target, TransTOI produces organization comparable to AF-TOI, consistent with induction deriving from the non-causal bidirectional structural signal rather than the specific BiLSTM topology. The causal Transformer target collapses in coverage without external alignment, supporting the causal LSTM as the most robust architecture for embedded deployment.

Post-hoc calibration does not restructure temporal entropy dynamics (seed 42; H_{pre} consistent across seeds 42–45)

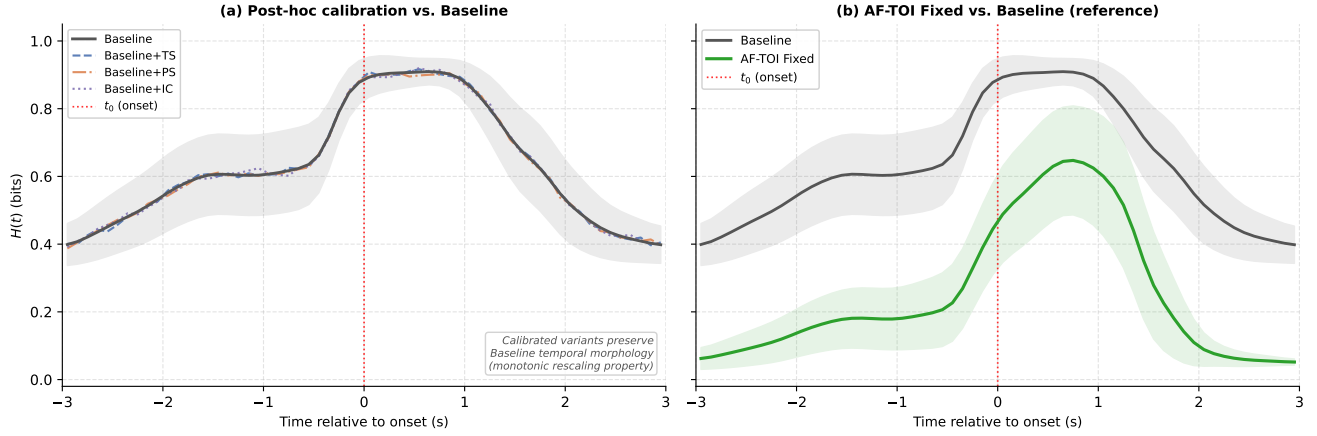


Figure 12: Mean entropy trajectory $H(t)$ around onset for seed 42; aggregate H_{pre} metrics are consistent across seeds 42–45. Panel (a) shows post-hoc calibration variants; panel (b) shows AF-TOI.

Table 14

Epistemological control via post-hoc calibration applied to Baseline with and without MHEG. MHEG thresholds are re-calibrated per seed for each variant. ECE (cal): post-calibration ECE; ΔECE : improvement vs. uncalibrated Baseline (≈ 0.035). Reference: Baseline Fixed FR = 0.040 ± 0.052 , $TTD_{ef} = 0.602 \pm 0.608$ s; AF-TOI+MHEG FR = 0.004 ± 0.008 , $TTD_{ef} = 0.101 \pm 0.155$ s, 48.7% computation suppression.

Configuration	ECE (cal)↓	ΔECE ↑	F1↑	FR↓	TTD_b (s)↓	TTD_{ef} (s)↓	%Sup↑
Baseline+TS	0.0290	0.0059	0.4410	0.0104	0.1283	0.2324	0.0%
Baseline+TS+MHEG	0.0326	0.0062	0.4835	0.0250	0.1877	0.4377	58.8450%
Baseline+PS	0.0184	0.0165	0.4687	0.0000	0.0760	0.0760	0.0%
Baseline+PS+MHEG	0.0203	0.0170	0.4608	0.0021	0.1013	0.1221	62.0800%
Baseline+IC	0.0075	0.0274	0.5042	0.0000	0.0362	0.0362	0.0%
Baseline+IC+MHEG	0.0110	0.0254	0.5040	0.0000	0.0527	0.0527	57.1100%

TTD_b : gross latency over detected episodes (partial). TTD_{ef} : effective detection delay with miss penalty. ΔECE : positive indicates improvement. %Sup: computation suppression rate by MHEG. Lower TTD_{ef} for PS and IC variants reflects threshold recalibration to the redistributed probability scale, not temporal trajectory reorganization; H_{strat} remains within the Baseline range for all three techniques.

Table 15

Non-causal reference architectural ablation (simplified protocol, 60 epochs, no 3-phase scheduler). Values not comparable with main study.

Causal Target Arch.	Non-Causal Reference Arch.	Approach	FR↓	TTD_{ef} (s)↓
Causal LSTM	None	Baseline Fixed	0.713 ± 0.069	7.223 ± 0.763
Causal LSTM	BiLSTM arch.	AF-TOI method	0.275 ± 0.189	3.089 ± 1.837
Causal LSTM	Transformer	TransTOI method	0.169 ± 0.163	2.022 ± 1.860
Causal Transformer	None	Baseline Fixed	0.958 ± 0.030	—
Causal Transformer	BiLSTM	AF-TOI method	0.467 ± 0.414	—

Absolute FR values exceed the main factorial study because the scheduler was omitted to isolate the reference architecture effect. TTD_{ef} omitted for Causal Transformer configurations due to near-complete coverage failure (FR ≈ 0.96), rendering the metric uninformative. This reinforces the scheduler's importance for transferred organization quality, without invalidating the non-causal bidirectional structural signal principle.

4.10. Negative Controls and Cost-Matched Robustness

To probe the central claim beyond the 2×2 factorial, we ran a complementary controlled campaign over nine onset/noise scenarios (Table 16), each with 360 episodes across seeds 42–45. The protocol adds two instruments not

used in the main study. First, a *compute-rate-matched* budget forces every negative control to spend the same fraction of frames on recurrent updates as MHEG ($\pm 2\%$), so that any difference reflects *where* computation is spent, not *how much*. Second, a near-onset false-suppression metric ($FS@onset$, the fraction of onset-window frames whose update was suppressed) isolates the specific failure that unsafe scheduling

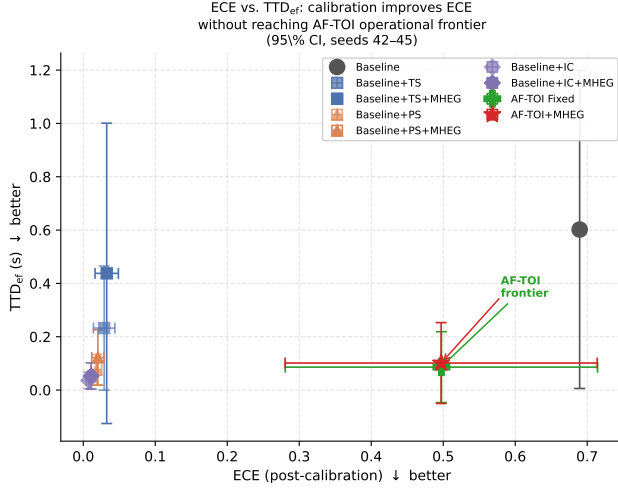


Figure 13: ECE– TTD_{ef} operational locus (CI 95%, seeds 42–45) for Baseline, post-hoc calibration variants, and AF-TOI.

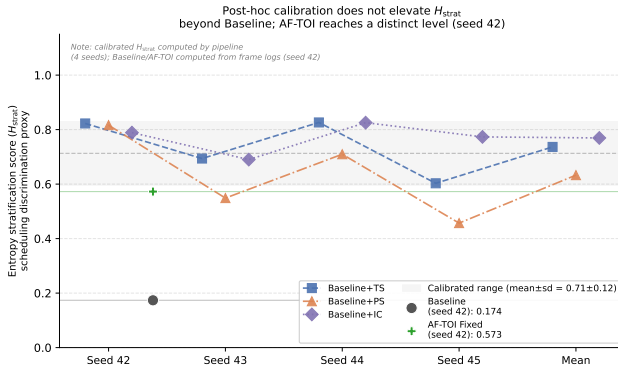


Figure 14: Pre-onset entropy stratification score per seed for seeds 42–45, comparing Baseline, post-hoc calibration variants, and AF-TOI.

produces. Coverage and miss rate are reported as percentages; compute rate is the fraction of frames on which the update runs. These magnitudes are on the stress-protocol scale and are not comparable to the aggregate TTD_{ef} of Section 4.4; the study is designed to rank safety at fixed cost, not to re-estimate efficiency. The budget-matched controls used here (random, unstructured, and lagged allocators) are intentionally harsher than the Baseline+MHEG negative control of Section 4.4: they hold the compute budget fixed and strip temporal structure, isolating the effect of *where* computation is placed rather than how much.

Cost-matched safety. The decisive result is that structure, not compute budget, drives safety. Table 17 matches every control to the MHEG budget within $\pm 2\%$. At *identical* cost, MHEG preserves full coverage with zero onset-window suppression, while budget-matched controls degrade near onset: under abrupt onsets the random and unstructured controls reach 41–44% FS@onset, and under rare transient events coverage collapses to 22–24% (miss rate $\geq 75\%$) at

Table 16

Controlled robustness scenarios (each 360 episodes, seeds 42–45).

Scenario	Onset	Entropy	Stressed failure mode
balanced_onsets	mixed	org.	cost saving (reference)
abrupt_onset_dominant	abrupt	org.	false suppression at abrupt onset
progressive_onset_dominant	progressive	org.	delayed detection (TTD_{ef})
rare_abrupt_critical	abrupt	org.	missed transient event (coverage)
high_noise_entropy	mixed	degr.	entropy scheduler misfires
kinematic_signal_missing	abrupt	org.	loss of kinematic protection
entropy_lag	abrupt	degr.	lagged entropy misses onset
false_onset	mixed	org.	wasted compute / false trigger
no_onset_negative_control	mixed	org.	sanity: FS must be ≈ 0

the same budget. Spending the same computation without temporal structure is therefore unsafe.

Negative-control sweep. Across all nine scenarios, MHEG (hybrid and calibrated) held FS@onset at 0.0 wherever the kinematic channel was available, whereas the budget-matched unstructured (NC2/NC6), random (NC1), and lagged (NC3) controls incurred near-onset suppression on every abrupt-dominated scenario, peaking at 56.7% FS@onset for the lagged control under abrupt onsets. The no_onset_negative_control sanity check behaved as required, with false suppression ≈ 0 for all methods. When the kinematic channel was removed (kinematic_signal_missing), MHEG retained coverage 100% with only 11.5% FS@onset, while the matched controls became unsafe (33–46% FS@onset), confirming the kinematic trigger as the intended fallback for sub-onset windows.

Entropy-lag sensitivity. Table 18 varies the entropic-signal lag. Progressive onsets tolerate lag (coverage stays 100% through 10 frames), but abrupt and especially rare transient onsets degrade sharply: at a 5-frame lag the rare-transient scenario loses all coverage (miss 100%). This is the mechanistic signature of the paper’s thesis, namely that near-onset entropy timing, not aggregate calibration, governs safe suppression.

Ablation of the scheduling channels. A per-channel ablation confirms that both components are necessary. Entropy-only scheduling preserves coverage but adds delay (TTD_{ef} rises relative to the hybrid), while kinematic-only scheduling is catastrophic on progressive onsets, collapsing coverage to 0% because slow transitions never trigger the kinematic gate. The hybrid matches the always-compute delay at a fraction of the cost, and paired tests over seeds rank the hybrid strictly above the entropy-only control on false suppression

Table 17

Cost-matched comparison ($\pm 2\%$ compute budget). MHEG vs. budget-matched negative controls. Coverage, miss and FS@onset in percent; compute rate is the fraction of frames updated.

Scenario	Method	Compute rate	Coverage	Miss	FS@onset
balanced_onsets	MHEG (hybrid)	0.217	100.0	0.0	0.0
	NC1 random budget-matched	0.218	99.7	0.3	23.9
	NC2 unstructured aggressive	0.218	100.0	0.0	20.3
	NC6 cost-matched unstructured	0.218	100.0	0.0	20.3
abrupt_onset_dominant	MHEG (hybrid)	0.193	100.0	0.0	0.0
	NC1 random budget-matched	0.195	99.7	0.3	43.9
	NC2 unstructured aggressive	0.196	99.7	0.3	41.6
	NC6 cost-matched unstructured	0.196	99.7	0.3	41.6
rare_abrupt_critical	MHEG (hybrid)	0.066	100.0	0.0	0.0
	NC1 random budget-matched	0.066	21.9	78.1	86.2
	NC2 unstructured aggressive	0.067	24.4	75.6	85.1
	NC6 cost-matched unstructured	0.067	24.4	75.6	85.1

Compute rate is the per-scenario fraction of frames updated under the matched-budget protocol; it is deliberately distinct from, and not comparable to, the aggregate suppression rate (%Sup) of Section 4.4. Within each scenario the budget is matched *across methods* to isolate allocation quality at equal cost.

Table 18

Entropy-lag sensitivity by onset type. Coverage, miss and FS@onset in percent.

Lag	Scenario	Coverage	Miss	FS@onset
1	abrupt_onset_dominant	100.0	0.0	28.3
1	rare_abrupt_critical	100.0	0.0	26.3
3	abrupt_onset_dominant	100.0	0.0	56.7
3	rare_abrupt_critical	64.4	35.6	78.8
5	abrupt_onset_dominant	100.0	0.0	85.0
5	rare_abrupt_critical	0.0	100.0	100.0
10	rare_abrupt_critical	0.0	100.0	100.0

(Cliff's $\delta = -0.5$, large) with no delay penalty against always-compute ($\delta = 0$, negligible).

4.11. Embedded Deployment Viability

At runtime, the active infrastructure is limited to the causal LSTM target, the MHEG scheduler, constant thresholds, and the recurrent buffer (Table 19).

AF-TOI+MHEG reaches 48.7% computation suppression (%Sup) and 53.4% effective cost reduction (ms/f, per-frame latency relative to Baseline Fixed). Together, these metrics summarize the runtime economy of the deployed system.

Estimates for dedicated embedded platforms come from scaling projections; physical chip validation is required before production.

Experimental evaluation synthesis Across the six movements, the same result appears repeatedly: in this embedded causal setting, selective inference preserves coverage only when uncertainty is temporally differentiated around risk emergence.

AF-TOI induces the differentiation that makes MHEG's criterion discriminative, so computation goes to emergence windows and is withheld through redundant ones; the gain

Table 19

Embedded deployment viability indicators (unit-batch mode).

Indicator	Baseline	AF-TOI	AF-TOI+MHEG
Parameters		≈ 58 K	
FP32 / INT8		≈ 228 KB / ≈ 57 KB	
Recurrent buffer		1024 bytes	
FLOPs/timestep		≈ 28 KFLOPs	
Latency (ms/f)	1.015	0.920	0.473
Computation suppression	0%	0%	48.7%

traces to a reorganized failure distribution rather than a lower mean, it holds across seeds, and it carries over to intermediate-risk episodes never seen in training.

All of this runs within ≈ 58 K parameters and 1,024 bytes of static RAM – a footprint that automotive-grade microcontrollers and low-power accelerators can carry.

Artifact availability Source code and TERA-Gen synthetic datasets will be made available as described in the Data Availability statement.

5. Discussion

Temporal organization as the enabling condition for edge resource allocation The negative control is the clearest result in the study. Baseline+MHEG keeps the miss rate roughly unchanged but does not produce a useful efficiency gain. This clarifies the role of AF-TOI: it is not that AF-TOI raises classification scores, but that it creates the conditions under which selective computation remains safe under the coverage constraint.

Without differentiated dynamics, efficiency and coverage do not merely fail to improve together – they compete,

because the scheduler trims apparent per-frame cost while the effective delay climbs.

Once the trajectories are organized, the trade-off becomes less severe (Movements 4.2 and 4.3): H_t becomes informative for the scheduler, concentrating computation at high-instability timesteps and reusing buffers through the stable ones.

The premise behind SkipRNN [6] and ACT [8] – that low uncertainty marks content safe to skip – holds in activity recognition and breaks in causal edge inference, and the negative control is where that break becomes measurable.

The runtime allocator is therefore effective only when the model already carries temporal differentiation in its trajectories.

AF-TOI and scheduler reliability A model with 95% classification accuracy and an unsafe scheduler is not equivalent to one whose scheduler allocates correctly at emergence windows.

TERA shows that scheduler quality is at least as important as classification quality, and that protocols optimizing only accuracy or calibration do not guarantee the trajectory structure scheduling reliability depends on, a direct implication for edge-native ITS design.

Probabilistic calibration and temporal organization as independent edge design properties The calibration control separates two design properties. Temperature, Platt, and isotonic scaling improve aggregate alignment without touching weights or internal trajectories, and the $H(t)$ morphology around onset stays within the Baseline range for all three, confirming that monotonic rescaling does not induce temporal differentiation.

Where PS and IC reach lower TTD_{ef} under MHEG, it is threshold recalibration to the redistributed scale, not organizational restructuring, as the invariant H_{strat} profile shows. The frontier reachable through temporal induction stays qualitatively distinct from what scale adjustment can reach.

Marginal calibration and temporally conditioned operational reliability therefore vary independently and come from different mechanisms.

Effective coverage versus gross delay The Baseline shows lower gross delay on detected episodes, but that metric omits misses. For safety-critical edge systems, coverage across all episodes matters more than mean latency on successes, and any metric excluding misses reads optimistically.

Complementary role of the kinematic component When pre-onset windows fall below 0.6 s, temporal differentiation may not appear in entropy fast enough. MHEG's kinematic component covers this by triggering allocation unconditionally when instantaneous physical variation signals a sudden state change.

Using both components allows MHEG to handle gradual and abrupt emergency regimes without depending exclusively on entropy, which is important under abrupt onset.

Implications for edge AI deployment and vehicular edge integration The deployment profile follows mainly from the offline/online separation.

The computationally intensive stages – TERA-Gen, the BiLSTM reference, AF-TOI, and threshold calibration – remain in the development environment; the device runs only the causal LSTM, the calibrated thresholds, and the recurrent buffer, with no reference model to execute and no on-line retraining.

MHEG's own overhead is $\mathcal{O}(1)$ per timestep, negligible beside the recurrent update it gates.

At ≈ 58 K parameters, a 1,024-byte buffer, and ≈ 28 KFLOPs per timestep, the deployed model remains within the operating range of automotive-grade microcontrollers and accelerators.

Because the only artifacts that move to the device are calibrated weights and thresholds, it is compatible with OTA updates and needs no connectivity at inference time.

The ℓ_t signal, produced at $\mathcal{O}(1)$ cost, is the interface through which higher-level VEC and MEC coordination could later consume the node state – but that contact is an implication we develop in Section 6, not something evaluated here.

Future directions Three extensions are immediate. First, the MHEG scheduling signal could feed future VEC and MEC offloading policies (Section 6), triggering remote inference when local entropy exceeds a coordination threshold while suppressing redundant timesteps and reallocating the saved energy.

Second, online adaptation of MHEG thresholds under distributional shift is the main step needed before production deployment.

Third, adversarial robustness under sensor faults, malicious perturbations, and modality degradation is a natural safety extension.

Scope and applicability The campaign is scoped to a single embedded node running continuous causal inference, which isolates the temporal-organization effect from network-level confounders. No networked coordination is evaluated.

The mechanism could be reused in future distributed deployments; Section 6 discusses how the MHEG entropy gate could inform VEC and MEC offloading, with empirical validation under real network conditions left as future work.

Robustness against adversarial attacks, sensor faults, and cybersecurity threats is out of scope and noted in Section 5.

Methodological rationale for synthetic evaluation TERA-Gen is used here for a specific methodological reason. Controlled evaluation of temporal organization needs three properties naturalistic datasets cannot provide together at the required granularity: (i) *deterministic onset annotation*, removing the ≈ 0.31 s variability that would make scheduler differences indistinguishable at 9.6 fps; (ii) *causal isolation*, preventing correlated naturalistic confounders from masking

the effect of trajectory structure on selectivity; and (iii) *frame-precise attribution*, enabling exact quantification of when the scheduler allocates relative to onset.

TERA-Gen anchors synthetic trajectories to SHRP2 odds ratios [13], preserving statistical realism while achieving annotation precision naturalistic collection cannot guarantee.

It is not intended to replace naturalistic evaluation. Its role is to make the hypothesis testable at the required temporal granularity. Generalization to naturalistic data is the primary direction for future validation (Limitation 1).

Threats to validity and deployment limitations The claims are bounded by four limitations.

First, evaluation relies on TERA-Gen synthetic data, designed to isolate the causal relationship between temporal organization and trustworthy selective computation under controlled onset; confirmation on naturalistic datasets such as DMD [2] and DADA [14] remains necessary for external validity.

Second, MHEG thresholds are calibrated on synthetic data and may need re-calibration under real-world shift from driver variability, illumination, and sensor noise.

Third, the factorial analysis uses four seeds (42–45); with $n = 4$, the Wilcoxon signed-rank test has a minimum one-sided p -value of 0.0625, which prevents conventional significance at the 0.05 threshold, so the Wilcoxon and Cliff's delta estimates are read as descriptive effect-size characterizations rather than definitive inferential tests.

The strongest evidence is therefore the consistent direction of the negative control, where MHEG on unorganized trajectories worsens TTD_{ef} while the same scheduler on organized trajectories preserves coverage and cuts computation.

Expanding to 10–12 seeds with an a priori power analysis is a priority for the next iteration.

Fourth, deployment feasibility is assessed through profiling and analytical estimation rather than execution on physical automotive hardware, so latency, energy, thermal behavior, and sensor-noise robustness remain to be characterized on STM32H7-, Cortex-M55-, and SA8155P-class platforms.

Planned real-data validation The primary next step is to test whether the temporal organization induced on TERA-Gen transfers to real driving video, using the DADA2000 dataset [14], which provides frame-level accident annotations – an abnormal-onset frame, an accident frame, and an abnormal-end frame – together with annotated contextual attributes (weather, lighting, scene, and road geometry).

The abnormal-onset frame serves as a real, frame-precise onset t_0 , and the interval up to the accident frame is the emergence window in which detection must occur; this is the timing reference that many naturalistic driver datasets lack and that TERA-Gen supplies synthetically.

The scene and contextual attributes map onto the visual and contextual channels of our feature schema.

Because DADA2000 is scene-facing and carries no ego-vehicle kinematics, the kinematic trigger of MHEG has no native input; we will therefore evaluate the entropic gate

both on its own and with an optical-flow-based kinematic proxy, an ablation that isolates the contribution of temporal organization from that of the kinematic trigger.

Evaluation will use clip-level splits stratified by scene and accident type, with clips from the same source kept on the same side to prevent leakage, and MHEG thresholds re-calibrated per split on real validation data and never transferred across domains.

We will compare three transfer regimes – zero-shot, with the organization-bearing recurrent layers frozen and only the input projection adapted; light fine-tuning, with the recurrent layers frozen and the input and output layers adapted; and full re-induction of AF-TOI on the real training split – against the same negative control used here (Baseline+MHEG).

Success is defined not by matching the synthetic magnitudes, which are on a different scale, but by reproducing the qualitative relationships established on TERA-Gen: the negative control failing to improve coverage, and AF-TOI+MHEG preserving coverage while suppressing computation.

On the real data we will report TTD_{ef} , FR, %Sup, and near-onset false suppression (FS@onset, Section 4.10) with the dataset's own episode length and sample size, alongside the accident-anticipation metrics standard for that dataset.

No real-data results are reported in this paper; this paragraph states the planned protocol only, and the entropy-lag tolerance measured in Section 4.10 sets the expectation that the scheduler should absorb the timing imprecision of real annotations up to a bounded lag.

6. Architectural Implications for Vehicular Edge Computing and Future Offloading

This section should be read as an architectural interpretation, not as an evaluation of a distributed system.

The campaign in Section 4 ran on a single embedded node under a fixed budget: no networked coordination, offloading, communication latency, or bandwidth analysis was evaluated.

The validated contribution remains single-node embedded selective computation.

The goal is narrower: to indicate how the validated node-level state signal ℓ_t could be used in future VEC and MEC architectures.

6.1. TERA as a Source of Edge Operational State Signals

The MHEG runtime output is a discrete signal $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$, emitted once per timestep at $\mathcal{O}(1)$ cost (Algorithm 1, Eq. 13).

It is compact, a two-bit integer that can be stored, transmitted, or logged without reference to LSTM weights, recurrent state, or sensor frames, and it is interpretable through the epidemiological calibration of TERA-Gen (Table 3).

When MHEG suppresses the update, ℓ_t comes from the buffered p_{t-1} with no inference at all.

The difference between ℓ_t and a thresholded classifier output is its temporal grounding. Under AF-TOI, ℓ_t reflects

phase-conditioned trajectory structure rather than instantaneous confidence, and the negative control of Section 4.4 shows a Baseline scheduler without this structure degrades TTD_{ef} by 11.8%.

The contribution here is the production of ℓ_t with experimentally supported temporal grounding. Its external use remains a downstream question.

6.2. Candidate Integration Paths for VEC and MEC Offloading

Resource allocation at the embedded tier The most immediate implication is within the on-board unit. Under AF-TOI+MHEG, 48.7% of frames are suppressed without violating $FR \leq 0.05$, freeing cycles that could support other on-board functions. Whether such sharing is feasible depends on the hardware platform and RTOS scheduler, which this work does not evaluate.

Compatibility with VEC and MEC offloading In VEC and MEC architectures, the on-board unit decides which tasks to keep local and which to offload to roadside or cloud infrastructure. The signal ℓ_t could serve as one urgency input among others, with persistent Alert or Critical indicating time-critical conditions and Normal or Attention indicating deferrable ones.

Such a policy could transmit only the compact symbol ℓ_t plus a summary feature window rather than raw frames, consistent with bandwidth-constrained partial offloading [53]. The activation fractions in Movement 6 (Section 4.7) are episode-level severity indicators, not measured communication rates; translating them into frame-level communication load requires network-level validation not performed here.

From scheduling signal to coordination primitive. In the borderline setting of Movement 6, activation fraction was more informative than instantaneous mean probability for summarizing the response to intermediate-risk episodes. A future policy could escalate on the persistence of elevated levels over a short window rather than on single readings, expressible as a rolling escalation statistic,

$$\Phi_t(w) = \frac{1}{w} \sum_{i=0}^{w-1} \mathbf{1}[\ell_{t-i} \geq 2], \quad (16)$$

the fraction of the last w timesteps at Alert or Critical.

$\Phi_t(w)$ adds only a bounded moving sum over a symbol the system already emits, at a memory cost of w two-bit symbols, negligible next to the 1,024-byte buffer.

It is not evaluated here but offered as a candidate bridge between the validated single-node mechanism and a future multi-tier policy, consistent with distributed early-exit designs [54] and VEC offloading frameworks [53, 55]; a future system could combine it with channel quality, queue length, and latency budget.

The single-node mechanism validated here can therefore be treated as a candidate coordination primitive for future VEC and MEC deployments, and the finding that activation

frequency summarized escalating risk more stably than instantaneous mean probability could inform how a future escalation policy is designed. Empirical validation under real network conditions remains future work.

7. Conclusion

This paper tested whether selective computation can remain safe under strictly causal vehicular and ad hoc edge inference, and what that safety depends on.

Our hypothesis was that it depends less on the scheduler's policy than on the temporal organization of the model's predictive trajectories – the property that makes uncertainty informative in the seconds around a critical transition.

Within the controlled setting studied here, the evidence supports this hypothesis, and the negative control is what makes it credible: with the scheduler applied to unorganized trajectories, TTD_{ef} worsened by 11.8% and coverage did not improve, so activating a scheduler is not, by itself, enough.

Given AF-TOI-organized trajectories, the same MHEG scheduler reached 53.4% effective cost reduction with TTD_{ef} of 0.101 s against 0.673 s.

The post-hoc calibration control ruled out the competing explanation: probabilistic rescaling improved ECE without reorganizing entropy in time, so marginal calibration and temporally conditioned reliability are separate properties with separate origins.

The result is not limited to driver monitoring. A training protocol that optimizes only classification accuracy or aggregate calibration can still leave a model unsafe to schedule on, because neither objective constrains the internal trajectory structure a causal scheduler reads.

On this evidence, temporal organization is worth treating as a design property in its own right for resource-constrained edge AI, rather than as an accidental byproduct of training.

The pipeline ends in the operational-risk signal $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$, produced at $\mathcal{O}(1)$ cost by the chain *Temporal Organization* $\rightarrow$ *Entropy Stratification* $\rightarrow$ *MHEG*.

Because it is compact, real-time, and interpretable at the level of the trajectory (Section 6.1), it is a plausible primitive for distributed and vehicular edge architectures to consume without touching the on-board loop – though we present that use as a design path (Section 6), not a validated result.

The validation scope is deliberately narrow. TERA is validated as a temporal-organization and scheduling substrate for a *single* embedded node, under strictly causal inference, on controlled synthetic data with a deterministic onset; the whole deployed workflow fits in ≈ 58 K parameters and 1,024 bytes of recurrent buffer, within reach of automotive-grade platforms.

It is not validated as a standalone detection system, and no networked coordination, offloading, or multi-node behavior was evaluated.

The VEC and MEC directions of Section 6 are implications of the ℓ_t signal, to be tested rather than claimed.

Accordingly, the immediate priorities are external validity on naturalistic data, robustness under distributional shift,

sensor faults, and adversarial perturbation, characterization on physical automotive hardware, and an empirical multi-node study of the offloading policies sketched here – each building on the single-node, entropy-gated mechanism this paper establishes.

Authors' Contribution Statement (CRediT)

Ângela Brígida Albarello: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing (original draft, review and editing). **Edna Dias Canedo:** Academic Supervision, Conceptualization, Writing (review and editing). **Vinícius P. Gonçalves:** Writing (review and editing). **Leandro A. Villas:** Supervision, Writing (review and editing). **Rodolfo I. Meneguette:** Writing (review and editing). **Geraldo P. Rocha Filho:** General Supervision, Writing (review and editing).

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Data Availability

The source code, synthetic datasets, configuration files, and evaluation scripts will be made publicly available upon publication. For peer review, an anonymized repository will be made available to the reviewers upon request.

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